PANJABUNIVERSITY,CHANDIGARH-160014 (INDIA)

(Estd.underthePanjabUniversityActVIIof1947 –enactedbytheGovt.ofIndia)

SYLLABI

Course Outline for
B.Voc. (Medical Laboratory and Medical Diagnostic Technology)

as per NEP 2020

W.e.f. session 2026-27

Course Outline for B.Voc (MLMDT)
as per NEP 2020

Semester-I

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
MLMDT 101	Core Discipline	Basics of Human Anatomy	2	2	4	40	10	40	10	100
MLMDT 102	Core Discipline	Introduction to Laboratory Equipments	2	2	4	40	10	40	10	100
MLMDT 103	Core Discipline	Introduction to Hematology	2	2	4	40	10	40	10	100
AEC / MIL	Ability Enhancement Course	English			2	40	10	-	-	50
SEC I	Skill Enhancement Course SEC 106	Basics of Enxcel			3	60	15	-	-	75
MDSC I	Multidisciplinary Course MDSC 101	Fundamental of Information Technology			3	60	15	-	-	75
VAC I	Value-Added Course VAC 103	Nutrition Fitness and Community Health			2	40	10	-	-	50
PW I	Project Work	Project Work-I& Viva-Voce	-	2	2	-	-	50	-	50
Total Credits					**24**					**600**

Semester-II

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
MLMDT 201	Core Discipline	General Pathology	2	2	4	40	10	40	10	100
MLMDT 202	Core Discipline	Basics of Biochemistry	2	2	4	40	10	40	10	100
MLMDT 203	Core Discipline	Analytical Laboratory Testing Process	2	2	4	40	10	40	10	100
AEC / MIL	Ability Enhancement Course	English			2	40	10			50
SEC II	Skill Enhancement Course	SEC options			3	60	15			75
IDC II	Multi disciplinary Course	Soft skills and Pjersonality Development			3	60	15			75
SIT 201*	Internship	Summer Industrial Training (SIT)*	-	4	4	-	-	100	-	100
Total Credits					**24**					**600**

Job Role - Medical Lab Technologist

*SummerIndustrialTrainingof2-4weeksinarelevantIndustryafter2ndSemesterExaminations during summer break. Training report by the student to be submitted within oneweek of start of 3rd Semester. Viva-Voce examination to be held within 3weeks of the start of 3rdsemester.

** Minimum Credits required to get Certificate Course in B. Voc. (MLMDT) I = 48 + 4 #.

#Exit 1: Certificate Course-Student can exit after completion of 1st year having secured 48 Credits and will be awarded a UG Certificate provided they complete a vocational course/internship of 4 credits during the summer vacation of the first year, in addition to the 6 credits from skill enhancement courses earned during first and second semester.

Semester-III

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
MLMDT 301	Core Discipline	Clinical Parasitology	2	2	4	40	10	40	10	100
MLMDT 302	Core Discipline	Fundamental of Microbiology	2	2	4	40	10	40	10	100
MLMDT 303	Core Discipline	Bacteriology, Virology and Mycology	2	2	4	40	10	40	10	100
AEC / MIL	Ability Enhancement Course	English			2	40	10	-	-	50
SEC III	Skill Enhancement Course SEC 301	Analytical skills on Benchtop Spectroscopy			3	60	15	-	-	75
IDC III	Multidisciplinary Course MDSC 301	Enterpreneurship Development			3	60	15			75
VAC II	Value-Added Course VAC 301	Business Ethics			2	40	10	-	-	50
PW II	Project Work	Project Work-II& Viva-Voce	-	2	2	-	-	50	-	50
Total Credits					**24**					**600**

Semester-IV

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
MLMDT401	Core Discipline	Clinical Laboratory Management	4	-	4	80	20	-	-	100
MLMDT402	Core Discipline	Histopathology, Cytology techniques and coagulation studies	2	2	4	40	10	40	10	100
MLMDT403	Core Discipline	Endocrinology	2	2	4	40	10	40	10	100
MLMDT404	Core Discipline	Microbiology -I	2	2	4	40	10	40	10	100
AEC / MIL	Ability Enhancement Course	English/MIL			2	40	10	-	-	50
VAC III	Value-Added Course	VAC options			2	40	10	-	-	50
SIT 401	Internship	Summer Industrial Training (SIT)*	-	4	4	-	-	100	-	100
Total Credits					**24**					**600**

Job Role – Health care service providers and Testing specialist

*SummerIndustrialTrainingof2-4weeksinarelevantIndustryafter4th SemesterExaminations during summer break. Training report by the student to be submitted within oneweek of start of 3rd Semester. Viva-Voce examination to be held within 3weeks of the start of 3rdsemester.

**Minimum Credits Required to get Diploma Course in B. Voc. (MLMDT) II = 96 + 4 #.

#Exit 2: Diploma Course-Student can exit after completion of 2nd year having secured 96 Credits and will be awarded a UG Diploma provided they complete a vocational course/internship of 4 credits during the summer vacation of the first year, in addition to the 9 credits from skill enhancement courses earned during first, second and third semester.

Semester-V

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
MLMDT501	Core Discipline	Clinical Biochemistry	2	2	4	40	10	40	10	100
MLMDT502	Core Discipline	Basics of Molecular Biology	2	2	4	40	10	40	10	100
MLMDT503	Core Discipline	Immunology and serology	2	2	4	40	10	40	10	100
MLMDT504^	Core Discipline	Basics of rDNA technology	2	2	4*	40	10	40	10	100
MLMDT505^	Core Discipline	Introduction to Biochemical Techniques	2	2	4*	40	10	40	10	100
MLMDT506^	Core Discipline	Safe Laboratory Practices	4		4*	80	20	-	-	100
VAC IV	Value-Added Course	VAC options			2	40	10			50
PW III	Project Work	Project Work-III& Viva-Voce			2	40	10			50
Total Credits					**24**					**600**

^Any two in 5th Semester

Semester-VI

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
_____601	Core Discipline	Advanced Technique in Hematology, Cytology and Histopathology	2	2	4	40	10	40	10	100
_____602	Core Discipline	Tumor and Cancer marker	2	2	4	40	10	40	10	100
_____603	Core Discipline	Microbiology -II	2	2	4	40	10	40	10	100
_____604	Core Discipline	Advanced Molecular Biology	2	2	4	40	10	40	10	100
_____605^	Core Discipline	Metabolic Biochemistry	2	2	4*	40	10	40	10	100*
_____606^	Core Discipline	Bioinformatics	2	2	4*	40	10	40	10	100*
_____607^	Core Discipline	Sensitization to Blood Banking and Infection Control			4*	80	20	-	-	100*
Total Credits					**24**					**600**

^Any two in 6th Semester

Job Role – Medical and Molecular diagnostic Laboratory Analyst,

Semester-VII

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
_____701	Core Discipline	Advanced techniques in Bioinformatics	2	2	4	40	10	40	10	100
_____702	Core Discipline	Clinical Pathology	2	2	4	40	10	40	10	100
_____703	Core Discipline	Molecular Diagnostics	2	2	4	40	10	40	10	100
_____704^	Core Discipline	Biophysics			4*	80	20	-	-	100*
_____705^	Core Discipline	Advanced Molecular Biology	2	2	4*	40	10	40	10	100*
_____706^	Core Discipline	Introduction to Medical Technology Healthcare system			4*	80	20	-	-	100*
SIT 701	Internship	Summer Industrial Training (SIT)*	-	4	4	-	-	100	-	100
Total Credits					**24**					**600**

^Any two in 7th Semester

Semester-VIII

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
_____801	Core Discipline	Research methodology			4	80	20			100
_____802	Core Discipline	Biostatics			4	40	10	40	10	100
_____803	Core Discipline	Research work/ Industrial Training			16					400
_____804	Core Discipline	-	-	-	-	-	-	-	-	-
_____805^	Core Discipline	-	-	-	-	-	-	-	-	-
_____806^	Core Discipline	-	-	-	-	-	-	-	-	-
_____807^	Core Discipline	-	-	-	-	-	-	-	-	-
Total Credits					**24**					**600**

^Any two in 8th Semester

Job Role – Medical and Molecular diagnostic Laboratory testing Supervisor,

*SummerIndustrialTrainingof2-4weeksinarelevantIndustryafter6th SemesterExaminations during summer break. Training report by the student to be submitted within oneweek of start of 7th Semester. Viva-Voce examination to be held within 3weeks of the start of 7thsemester.

ESSENTIAL COMPONENTS OF COURSE OF B.VOC (MLMDT) (SEMESTERS I & II)

PROGRAMME OUTCOMES

PO1. Understanding the concept, principles of working of the Medical and Molecular Diagnostic lab**.**

PO2. Understanding the Human anatomy and various diseases related to human body**.**

PO3. Understanding the concept and procedures of various Analytical and Biochemical test**.**

PO4. To develop ability to work in the Hospital and understand the Healthcare system.

PO5. To develop awareness of the Marketing strategies, Management hierarchy involved in various labs

PO6. To develop the skill and application of Information Technology in the field of biosciences

Semester I

<table>
<tr><td>1</td><td>Course Title</td><td>Basics of Human Anatomy</td></tr>
<tr><td>2</td><td>CourseCode</td><td>Skill MLMDT 101</td></tr>
<tr><td>3</td><td>Typologyof Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration/ LP (Lectures)
(Practical)</td><td>60 hours of lectures for Theory + 30 hours of practicals
(1 credit = 15 hours of theory learning based on 3 hrs./week)
(1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)
3-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>5</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>6</td><td>Max. Marks and Time</td><td>
<table>
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<tr><td>40</td><td>10</td><td>40</td><td>10</td></tr>
</table>
100Marks (80TH/PR+20IA)
a. The exam of this course will be of 3hours duration.
b. There will be no objective type questions.
c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
</td></tr>
<tr><td>7</td><td>Job Role</td><td>Medical Lab Technologist.</td></tr>
<tr><td>8</td><td>Course Objectives</td><td>Basic understanding of organization of body cells, tissues, organs, organ systems, and glands in human body.</td></tr>
<tr><td>9</td><td>Course Outcomes</td><td>On successful completion of the course, the Students will be able to
CO1.Gain the sufficient knowledge of cellular structure and its working.
CO2. Understanding of tissue system, organs and organ system.
CO3.Learning the cellular communication through receptor studies.</td></tr>
<tr><td>10</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
<tr><td>11</td><td>Instructions for paper setters and candidates</td><td>1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.</td></tr>
</table>

12 Course Content

UNIT– I

- Basic unit of body - Cell
- The anatomical organization of body cells, tissues, organs, organ systems, membranes and glands in human body
- Introduction to different types of tissues: Anatomy, epithelial tissue, muscular tissue, nervous tissue Different types of organ
- Brief Introduction of different types of body fluids, secretions and excretions
- Skeletal system: Bones, joints and muscles

UNIT– II

Digestive Organs:

- Teeth
- Tongue
- Oral cavity
- Pharynx
- Oesophagus
- Stomach
- Small intestine
- Large intestine
- Liver, Pancreas and Spleen

UNIT– III

Respiratory Organs:

- Nasopharynx
- Oropharynx
- Larynx
- Trachea
- Bronchi
- Lungs
- Thoracic cavity
- Pleura and Pleural cavity

Circulation System: Structure of Heart and Brief introduction of main blood vessels

UNIT– IV

Reproductive Organs: Male and Female Gonads

Nervous system and Sense organs: Brief Introduction of Central Nervous System and Peripheral Nervous System Anatomy of Brain, Spinal Cord, Nerves, Eye, Ear,
Olfactory Receptors, Gustatory Receptors

Excretory Organs:

- Cortex and medulla of Kidney
- Urinary Bladder
- Urethra: Male, female

Practical:

1. Preparation and study of mitosis in onion root tips
2. To demonstrate the presence of Barr body from the oral mucosa of Human female
3. Study of various organ through charts and models.

13 Suggested Resources for Reading:

1 Ross and Wilson, *Anatomy and Physiology in Health & Illness*
2. Murgesh, N., *Basic Anatomy and Physiology*, Sathya Publishers
3. Pearce, E., *Anatomy and Physiology* for Nurses, Jaypee Brothers
4. O' Rahilly, R.R., *Basic Human Anatomy*: *Regional Study of Human Structure*, WB Saunders Co. Ltd.
5. Sears, W.G., Winwood, R.S. & Smile, J.L., *Anatomy and Physiology for Nurses* and *Students of Human Biology*, Hodder Arnold Note: Latest edition of text books may be used.

1	**Course Title**	**Introduction to Laboratory Equipments**
2	**CourseCode**	**MLMDT 102**
3	**Typologyof Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**60 hours of lectures for Theory + 30 hours of practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **3-groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	<table><tr><td>External Theory</td><td>Internal Theory</td><td>External Practical</td><td>Internal Practical</td></tr><tr><td>40</td><td>10</td><td>40</td><td>10</td></tr></table> 100Marks (80TH/PR+20IA) **a.** The exam of this course will be of 3hours duration. **b.** There will be no objective type questions. **c.** Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. **d.** Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	**Medical Lab Technologist.**
8	**Course Objectives**	It provides opportunity to perform hands on practical training and Laboratory equipment**.**
9	**Course Outcomes**	**On successful completion of the course, the Students will be able to** **CO1.** To gain broad understanding of care of laboratory glassware, equipment and instrument**.** **CO2.** To gain broad understanding of setting up, calibrating, operating, cleaning, maintaining, and troubleshooting of laboratory equipment used in quantitative or qualitative analysis. **CO3.** To Calibrate and Validate the Clinical Laboratory instruments and glass wares**.** **CO4.** To understand Microscopy, working principle, maintenance and applications of various types of microscopes
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	**1. The syllabus of this paper has been divided into FOUR units.** **2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** **3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.** **4. All questions carry equal marks.**

12 Course Content

UNIT– I

- Brief understanding of laboratory planning and laboratory key learning outcome operations
- Introduction to operation and safety precautions of common Laboratory Equipments
- Principles and working of laboratory instruments: Incubator, Hot Air Oven, Water Bath, Anaerobic Jar, Centrifuge, & pipettes, colorimeter, Neubauer's Chamber Setting up, calibrating and validating, operating, cleaning, maintaining,
- Autoclave – Its structure, functioning, control and indicator Micrometry
- Elementary Knowledge of Analytical Chemistry: Balances- monopan, two pan, top pan, pH meter, spectrophotometer, fluorimeter, flame photometer, ion selective electrodes, urinometer

UNIT– II

- Techniques of Disinfection
- Sterilization: Definition, Classification and General Principle of Sterilization, Sterilization Techniques, Sterilization of rubber goods, laboratory equipment & other instruments.
- The cleaning and maintenance procedures of the machine
- Identify the cause of errors or other problems or defects in Equipments
- Glassware – Description of Glassware, its use, handling, and cleaning of glass ware

UNIT– III

- To study the compound microscope and its parts
- Microscopy: Working principle, maintenance, and applications of various types of microscopes. Care and handling of various microscopes
- Dark ground microscope
- Polarizing microscope
- Phase contrast microscope
- Interference microscope
- U.V. light microscope
- Electron microscope

UNIT-IV

- Ultra-microtomy
- Importance of conservation of resources in laboratories
- Museum techniques
- Lab safety and Instrumentation

Practical:

1. Demonstration of the use of autoclave, centrifuge, and spectrophotometer
2. Cleaning and sterilization of glassware using hot air oven, autoclave etc.
3. Calibration of apparatus and glassware
4. Demonstration of various equipments for the estimation of hemoglobin, WBC, RBC (Neubauer chamber setting up)
5. Demonstration of parts of simple and compound microscope, phase contrast microscope, its functioning and care
6. Demonstration of safety rules in laboratory like proper handling of patients specimens and disposal of syringes, needles etc.

13 Suggested Resources for Reading

1. At the Helm: *A Laboratory Navigator*, Barker, K., Cold Spring Harber Laboratory Press
2. Basic Medical Laboratory Techniques, Estridge, B.H., Walters, N.J. & Reynold, A. Cengage Learning
3. Chatwal, G.R., & Anand, S., Instrumental Methods of Chemical Analysis, HpH
4. Roskans, J. & Rodgers, L., Lab Ref: *A Handbook of Recipes, Reagent and Other Reference Tools for Use at the Bench* , Cold Spring Harber Laboratory Press
5. Haminond , M., *Oxford Consice Medical Dictionary* , Oxford University Press

<table>
<tr><td>1</td><td>Course Title</td><td>Introduction to Hematology</td></tr>
<tr><td>2</td><td>CourseCode</td><td>MLMDT 103</td></tr>
<tr><td>3</td><td>Typologyof Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration/ LP (Lectures)
(Practical)</td><td>60 hours of lectures for Theory + 30 hours of practicals
(1 credit = 15 hours of theory learning based on 3 hrs./week)
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(1 Semester = 15 weeks)
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<tr><td>5</td><td>Credits</td><td>4 Credits</td></tr>
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100Marks (80TH/PR+20IA)
a. The exam of this course will be of 3hours duration.
b. There will be no objective type questions.
c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
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<tr><td>7</td><td>Job Role</td><td>Medical Lab Technologist.</td></tr>
<tr><td>8</td><td>Course Objectives</td><td>To gain understanding of blood and components of blood To gain knowledge of hematological Diseases and hematological investigations.</td></tr>
<tr><td>9</td><td>Course Outcomes</td><td>On successful completion of the course, the Students will be able to
CO1. Learn the various component of blood.
CO2. Understand the blood circulation, coagulation, and haemostatis.
CO3.Understand the various haemodynamic disorder.</td></tr>
<tr><td>10</td><td>Pedagogy of the Course Work</td><td>ClassRoomLecture, Laboratory,Interactivesessions,PPT,Practicaland GroupDiscussions,Case Studiesetc.</td></tr>
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12 Course Content

UNIT– I

- Introduction to hematology and laboratory organization. Composition and functions of blood and lymph
- Detailed study of Hemoglobin and functions of hemoglobin
- Blood groups including Rh. Factor
- Detailed study of Reticulocytes
- Formation of blood. Morphology of normal blood cells and their identifications

UNIT– II

- Hemostasis– Definition, Basic concept and principle, Basic steps involved in Hemostasis. Coagulation– Basic Physiology, coagulation factors, Fibrinolysis
- Various anticoagulants, their uses, mode of action and their merits and demerits. Normal and absolute values in hematology
- Quality assurance in hematology

UNIT– III

- Descriptive study of RBC abnormalities, Disorders related to RBC Normal white cell count & physiological variation, Normal white cell count & physiological variation. Hematological Diseases: Anemia and various types of anemia, Thalassemia, Polycythemia, hemolytic disease of new born, multiple myeloma.
- Leukemia: Definition and classification
- Introduction to Bleeding Disorders

UNIT-IV

- Laboratory preparation for coagulation tests, Routine coagulation tests, prothrombin time, plasma recalcification time, partial thromboplastin time, activated partial thromboplastin time, thrombin time.

Practical:

1. Cleaning of Laboratory glassware in haematology
2. Clinical significance, specimen collection, laboratory investigation & preservation of blood forvarious haematological investigations
3. Preparation of blood smears
4. Total leukocyte count and Differential leukocyte count
5. Determination of total erythrocyte (RBC) count and platelet counts. Errors involved and mean tominimize such errors
6. Haemoglobinometry, various methods of estimation of Hb, errors involved and standardization of instrument for adaptation for Hb estimation.
7. Romanowsky dyes, preparation and staining procedures of blood smears
8. Laboratory tests for assessing bleeding disorders
9. Determination of erythrocyte sedimentation rate (ESR)

13 Suggested Resources for Reading

1. *Atlas of Haematology*, McDonald, G.A., Cruickshank, B. & Paul, J. Churchill

2. *Clinical Haematology*, Ludlam, C.A., Churchill Living Stone, Living Stone

3. Bain, B.J., Bates, I., & Laffan, M.A., Dacie and Levis, *Practical Haematology* Elsevier

4. *Practical Haematology (International edition),* Dacie, J.V. & Lavie, S,M., Churchill Living Stone

5. *Haematology,* Besa, E.C., Lippincott, Williams & Wilkins

6. *Haematology* : Pathophysiological Basis for Clinical Diagnosis, Reich, P.R. & Robinson, S.H., Lippincott Williams & Wilkins

7. *Haematology for Students and Practitioners*, Sood, R., Jaypee Brothers Medical Publishers

8. *Hand Book of Medical Laboratory Technology*, V.H., CBJ

Semester II

<table>
<tr><td>1</td><td>Course Title</td><td>General Pathology</td></tr>
<tr><td>2</td><td>CourseCode</td><td>MLMDT 201</td></tr>
<tr><td>3</td><td>Typologyof Course</td><td>Core Discipline</td></tr>
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100Marks (80TH/PR+20IA)
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<tr><td>7</td><td>Job Role</td><td>Medical Lab technologist.</td></tr>
<tr><td>8</td><td>Course Objectives</td><td>Understanding the general principles, terminology, diagnostic procedure and basic concept of pathology.</td></tr>
<tr><td>9</td><td>Course Outcomes</td><td>On successful completion of the course, the Students will be able to
CO1. Study cell injury, death and adaptation .
CO2. Understand the concept of cancer and tumor.
CO3. Get knowledge of genetic and paediatric diseases.</td></tr>
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4. All questions carry equal marks.</td></tr>
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12 Course Content

UNIT– I

- Normal Cell
- Cell Injury: Types of cell injury, Etiology of cell injury, morphology of cell injury, Cellular swelling
- Cell death: Types- Autolysis, necrosis, apoptosis & gangrene
- Cellular adaptations: Atrophy, hypertrophy, hyperplasia & dysplasia
- Acute inflammation, Vascular events, cellular events, Inflammatory cells, Morphology of acute inflammation, Chronic inflammation
- Regeneration, repairs, wound healing and Healing in specialized tissues

UNIT– II

- Hemodynamic disorders: Internal environment, Normal water and electrolyte balance,Disturbances of body fluids and
- electrolytes, Oedema
- Disturbances in volume of circulating blood: Hyperemia and congestion, Hemorrhage and shock
- Circulatory disturbances of obstructive nature: Thrombosis, Ischemia, Infarction

UNIT– III

- Neoplasia: Nomenclature and classification, Characteristics of tumors, Local invasion and metastasis
- Prognostic markers, Grading and staging of cancer
- Viruses and tumor
- Clinical aspects of neoplasia, Tumor host interrelationship

UNIT-IV

- Diagnosis of cance
- Genetic and pediatric diseases: Introduction to Genetic diseases, Developmental defects and Cytogenetic abnormalities
- Nutritional diseases: Obesity, Starvation, Protein energy malnutrition, Disorders of vitamins and Trace element

Pathology

1. To study cultural isolation and identify of pus miscrobes from wound.
2. Urine Analysis
3. To identify pus microbes at genetic level
4. Study of various microscopes

13 Suggested Resources for Reading

1 Mohan, H., Textbook of Pathology, Jaypee Brothers Medical Publishers

2 Abbas, A.K., Andra, T.D., Astu, J.C. & Kumar, V., Basic Pathology

3 Cree, I.A., Pathology : A comprehensive Review, CRC Press

4 Strayer, D.S., & Saffitz, J.E., Rubin's Pathology: Mechanism of Human Disease, Lippincott, Williams & Williams

<table>
<tr><td>1</td><td>Course Title</td><td>Basics of Biochemistry</td></tr>
<tr><td>2</td><td>Course Code</td><td>MLMDT 202</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration/ LP (Lectures)
(Practical)</td><td>60 hours of lectures for Theory + 30 hours of practicals
(1 credit = 15 hours of theory learning based on 3 hrs./week)
(1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)
3-groups will be created for the purpose of Practical for each unit</td></tr>
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<tr><td>40</td><td>10</td><td>40</td><td>10</td></tr>
</table>
100Marks (80TH/PR+20IA)
a. The exam of this course will be of 3hours duration.
b. There will be no objective type questions.
c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
</td></tr>
<tr><td>7</td><td>Job Role</td><td>Medical Lab Technologist.</td></tr>
<tr><td>8</td><td>Course Objectives</td><td>To gain elementary knowledge of Biochemistry.</td></tr>
<tr><td>9</td><td>Course Outcomes</td><td>On successful completion of the course, the Students will be able to
CO1. Get knowledge of inorganic chemistry and preparation of solution and buffers.
CO2. Understand organic compound, their structure and properties.
CO3.Detailed knowledge of various biomolecules like Carbohydrates, Lipids, Proteins and Nucleic acid.</td></tr>
<tr><td>10</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
<tr><td>11</td><td>Instructions for paper setters and candidates</td><td>1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.</td></tr>
</table>

12 Course Content

UNIT– I

- Elementary Knowledge of Inorganic Chemistry: Role and properties of Water, Acids, bases and salts, Acid Base Balance, pH, Henderson- Hasselbalch equation.
- Normality, Molarity and Molality, Mole, Molar solutions, normal solutions, buffer solutions, physiological buffers, percent solutions, saturated solutions, standard solutions

UNIT– II

- Elementary Knowledge of Organic Chemistry: Organic compounds- Alcohol, Aldehyde, Ketone and Carboxylic acid: General Structure, Nomenclature, Classification and Physical Properties.
- Elementary Knowledge of Physical Chemistry- Osmosis, osmotic pressure, dialysis, surface tension, diffusion, hypotonic, hypertonic, and isotonic solutions. Definition and classification of some colloids and crystalloids

UNIT– III

- Carbohydrates: Definition, Classification, Biomedical importance & properties.
- Amino Acids and Proteins: Definition, Classification and Biomedical importance & properties.

UNIT-IV

- Lipids: Definition, Classification and Biomedical importance, Fatty liver, Cholesterol & its clinical significance, Lipoproteins in the blood & their functions.
- Introduction and general properties of Nucleic Acids Definition, Basic structure of DNA and RNA, types of RNA, Functions of nucleic acids.

Practical:

1. Preparation of buffer solutions and measurement of their pH
2. Verification of Beer Lambert's Law
3. To prepare the different concentrations of solutions
4. To find out the normality of the given solution
5. Qualitative tests for:
(a) Carbohydrates.
(b) Amino acids and proteins,
6. Estimation of ascorbic acid by dye method.
7. Estimation of Amino acids by ninhydrin method.
8. Estimation of Protein by biuret & lowry method.

13 Suggested Resources for Reading

1. A Manual of Laboratory & Diagnostic Tests, Frances Fischbach
2. Biochemistry, Voet and Voet
3. Biochemistry, Stryer
4. Biochemistry, U. Satyanarayan. & U. Chakrapan
5. Clinical Biochemistry, Richard Luxton
6. Clinical Diagnosis & Management by Laboratory Methods, John Bernard Henary
7. Clinical Biochemistry, G. Guru
8. Handbook of Biochemistry, M.A. Siddique
9. Textbook of Medical Biochemistry, S. Ramkrishnan
10. Biochemical Techniques, K. Choudhary
11. Text book of Medical Biochemistry, Chaterjee & Shinde
12. Principles of Biochemistry, David L. Nelson
13. Principles of Biochemistry, Lehninger
14. Textbook of Biochemistry and Human Biology, G.P. Talwar
15. Textbook of Medical Laboratory Technology, Godkar and Godkar
16. Outline of Biochemistry, Conn Stumpf
17. Principles of Internal Medicine, Issel bacher
18. Proteins and Proteomics : Laboratory Manual, Richard J. Simpson
19. Purifying Proteins for Proteomics: Laboratory Manual, Richard J. Simpson
20. Enzymes: Biochemistry, Biotechnology & Clinical chemistry, Palmer Trevor, Publisher: Horwood Pub. Co.
21. Outlines of Biochemistry, Erice Conn & Paul Stumpf; John Wiley and Sons
22. Fundamentals of Biochemistry, Donald Voet & Judith Voet , John Wiley and Sons, Inc
23. Lehninger, Principles of Biochemistry, David Nelson & Michael Cox, W.H. Freeman and company
24. Biochemistry, Jeremy Berg, Lubert Stryer, W.H. Freeman and company
25. Biochemical Methods for Agricultural Sciences – Sadasivam and Manikam. Wiley Eastern Limited
26. Practical Clinical Biochemistry Harold Varley, CBS

27. An Introduction to Practical Biochemistry, David T Plummer. Tata McGraw-Hill Publishing Company Limited,
28. Biometrics Identity, Sameer Nanawati
29. Medicinal Chemistry, Ashutosh Kar

<table>
<tr><td>1</td><td>Course Title</td><td>Analytical Laboratory Testing Process</td></tr>
<tr><td>2</td><td>Course Code</td><td>MLMDT 203</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration/ LP (Lectures) (Practical)</td><td>60 hours of lectures for Theory + 30 hours of practicals
(1 credit = 15 hours of theory learning based on 3 hrs./week)
(1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)
3-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>5</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>6</td><td>Max. Marks and Time</td><td>
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<tr><td>40</td><td>10</td><td>40</td><td>10</td></tr>
</table>
100Marks (80TH/PR+20IA)
a. The exam of this course will be of 3hours duration.
b. There will be no objective type questions.
c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
</td></tr>
<tr><td>7</td><td>Job Role</td><td>Medical Lab Technology.</td></tr>
<tr><td>8</td><td>Course Objectives</td><td>To gain broad understanding of chemicals/reagents and procedure useful in sample analysis.</td></tr>
<tr><td>9</td><td>Course Outcomes</td><td>On successful completion of the course, the Students will be able to
CO1.learn about collection and maintaining records of biological specimen and various body fluid sample.
CO2. gain broad knowledge of performing various routine haematological tests, urine tests, stool tests, semen tests and sputum tests..
CO3. Interpret the results of the above test for further diagnosis.</td></tr>
<tr><td>10</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
<tr><td>11</td><td>Instructions for paper setters and candidates</td><td>1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.</td></tr>
</table>

12 Course Content

UNIT– I

- Collection and recording of biological specimens. Collection, Handling and transportation of Blood Sample
- Routine Hematological Tests
- Examinations of sample for serum, Urea and Uric acid. Estimation of Essential electrolytes: Sodium, potassium, calcium, chloride and phosphorus
- Determination of blood sugar level of plasma (or serum): Ortho-toluidine method and Glucose oxidase method. Determination of serum urea nitrogen: Diacetyl monoxime method. Determination of serum creatinine: Alkaline picrate method. Determination of serum total cholesterol

UNIT– II

- Determination of serum bilirubin: Malloy and Evelyn method and DMSO method
- Laboratory investigation for anaemia
- Laboratory investigation for disseminated intravascular coagulation (DIC)
- Mechanism of fibrinolysis test for fibrinolysis. Platelet function tests and their interpretation
- Erythrocyte sedimentation rate (ESR), factors influencing ESR and various procedures for its estimation with their significance. Determination of pack cell volume (PCV). Physiological variations in Hb, PCV, TLC and platelets
- Determination and calculation of red blood indices: MCV, MCH, MCHC

UNIT– III

- Determination of absolute Eosinophil count and Reticulocyte count
- Automated systems in hematology
- Semen analysis: Composition of semen, Collection and preservation of semen, Physical and chemical examination of semen, microscopic examination of semen, Clinical significance of semen analysis
- Sputum analysis: Composition of sputum, Collection and preservation of sputum, Physical and chemical examination of sputum, microscopic examination of sputum, Clinical significance of sputum analysis

External Theory	Internal Theory	External Practical	Internal Practical
40	10	40	10

UNIT-IV

Urine Examination:

- Collection, Handling and transportation of Urine
- Urine analysis, routine examination of urine (physical examination of urine), rapid chemical testsof Urine
- Determination of specific gravity of urine by urinometer and refractometer
- Biochemical Test Profile (Quantitative determination of Urine): Amylase, Calcium, Chlorides, Creatinine, Sodium, Potassium, Glucose, Proteins, Urea nitrogen, uric acid

Practicals:

1. All Biochemical tests mentioned in Theory

13 Suggested Resources for Reading

1. Textbook of Medical Laboratory Technology, Godkar and Godkar
2. Research Methodology in Medical Sciences, Chandorkar
3. Practical Clinical Biochemistry, Harold Varley
4. Medical Laboratory Sciences, Theory & Practical, A. Kolhatkar
5. Medical Laboratory Technology – Volume I, Kanai Mukherjee
6. Medical Laboratory Technology – Volume II, Kanai Mukherjee
7. Medical Laboratory Technology Methods & Interpretation, Ramnik Sood
8. Enzymes: Biochemistry, Biotechnology & Clinical chemistry, Palmer Trevor, Publisher: Horwood Pub. Co.
9. Outlines of Biochemistry, Erice Conn & Paul Stumpf ; John Wiley and Sons
10. Fundamentals of Biochemistry, Donald Voet & Judith Voet , John Wiley and Sons, Inc
11. Lehninger, Principles of Biochemistry, David Nelson & Michael Cox, W.H. Freeman and company,
12. Biochemistry, Jeremy Berg, Lubert Stryer, W.H.Freeman and company
13. Biochemical Methods for Agricultural Sciences – Sadasivam and Manikam. Wiley Eastern Limited
14. Practical Clinical Biochemistry Harold Varley, CBS
15. An Introduction to Practical Biochemistry, David T Plummer. Tata McGraw-Hill Publishing Company Limited

1	Course Title	Clinical Parasitology
2	**CourseCode**	**Skill MLMDT 301**
3	**Typologyof Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**60 hours of lectures for Theory + 30 hours of practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **3-groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100Marks (80TH/PR+20IA) **e.** The exam of this course will be of 3hours duration. **f.** There will be no objective type questions. **g.** Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. **h.** Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	**Medical Lab Technologist.**
8	**Course Objectives**	**To gain an appreciation of the importance of parasitic infection in human health and disease.**.
9	**Course Outcomes**	**On successful completion of the course, the Students will be able to** **CO1.**Gain the sufficient knowledge of various parasites**.** **CO2**.Understanding of tissue, **organs affected by parasites, diseases caused by parasites.** **CO3 .**Learning the epidemiology, pathogencity , cure ,treatment of various diseases.
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	**5. The syllabus of this paper has been divided into FOUR units.** **6. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** **7. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.** **8. All questions carry equal marks.**

12 Course Content

UNIT– I

- Parasitology Definition - parasitism, Host, Vectors
- Classification of Parasites
- Protozoa: Intestinal Amoebae, E. Histolytica - Life cycle, Morphology, Disease& Lab Diagnosis.

UNIT– II

- Malarial Parasite, Plasmodium vivax- Life cycle, Morphology, disease & lab diagnosis .
- E. coli- Life cycle, Morphology, Disease & Lab Diagnosis.
- Differences between P. vivax and P. malaria.

UNIT– III

- Trichomonas vaginalis- Life cycle, Morphology, Disease& lab diagnosis.
- Flagellates of intestine/genitalia, Giardia lamblia &Lab Diagnosis.
- Trematoda: Fluke

UNIT– IV

- Nematodes: Intestinal Nematode- Ascaris, Life cycle, Morphology, disease & lab diagnosis.
- Tissue Nematode: W. Bancrofti, Life cycle, Morphology, disease & lab diagnosis
- Pin Worms: Life cycle, Morphology, disease & lab diagnosis.

Practicals:

1. Urine analysis
2. Detection of malarial parasite
3. Detection of trypanosomes (the causal agent of sleeping sickness)
4. Laboratory diagnosis of kala azar
5. Quantitative determination of serum (or plasma) IgG class antibodies to toxoplasma gondii by ELISA
6. Determination of IgM class antibodies to toxoplasma gondii by ELISA

13 Suggested Resources for Reading:

1. Kelkar, S.S. & Kelkar, R.S., A Textbook of Parasitology, Sangam Books Ltd.
2. Medical Parasitology, Karyakarte, R. & Daml;e, A., Booka & Allied Ltd.
3. Godkar,P.B. & Godkar, D.P., A.Text Book of Medical Laboratory Technology
4. Henry's, McPherson, R.A., & Pnius, M.R.., Clinical diagnosis and management by laboratory methods, Elsevier

1	**Course Title**	**Fundamentals of Microbiology**
2	**CourseCode**	**MLMDT 302**
3	**Typologyof Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**60 hours of lectures for Theory + 30 hours of practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **3-groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	

External Theory	InternalTheory	External Practical	Internal Practical
40	10	40	10

100Marks (80TH/PR+20IA)

e. The exam of this course will be of 3hours duration.
f. There will be no objective type questions.
g. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
h. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees

7	**Job Role**	**Medical Lab Technologist.**
8	**Course Objectives**	**To understand the role of parasites and vectors in disease transmission, and the most appropriate control strategie.**
9	**Course Outcomes**	**On successful completion of the course, the Students will be able to** **CO1.** To gain broad understanding of microbes and their morphology. **CO2.**To understand the role of bacteria in various fields. **CO3.** To study role of bacteriophages**.** **CO4.** To study in detail about fungi, algae and viruses.
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	**5. The syllabus of this paper has been divided into FOUR units.** **6. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** **7. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.** **8. All questions carry equal marks.**

12 Course Content

UNIT– I

- Introduction and brief history, development, scope and applications of Microbiology
- Growth and nutrition of microbes
- Biotransformation of Antibiotics
- Nature of Microbial World: Prokaryotes and eukaryotes, growth pattern in microbes

UNIT– II

- Morphology, General characteristics & Bacteriophages: Morphology, multiplication, detection and enumeration
- Organization of cell wall, cell membrane, flagella and capsules in bacteria
- Classification of bacteria
- Fine structure of bacteria

UNIT– III

- Morphology, Pathogenicity and laboratory diagnosis of human viruses
- Classification and general properties of Viruses
- Morphology, General characteristics and classification of pathogenic fungi
- General characteristics of actinomycete and algae

UNIT-IV

- Methods of Microbiology: Isolation of pure cultures, theory and practice of sterilization
- Staining of microbes, Theory of Gram staining
- Preparation, uses and standardization of culture media
- Principles of staining methods and preparation of reagents
- Principles and methods of sterilization
- Uses and mode of action of antiseptics and disinfectants

Practical:

1. To perform Simple and Gram staining
2. Preparation of media and culture plates.
3. Pure culture techniques for isolation of micro-organism
4. To perform antibiotic sensitivity test.
5. Check the efficacy of disinfectant.
6. Preparation of culture media, spread plates, pour plates selective media, differential media.
7. Separation of pure cultures and study the effect of selective nutrients on prokaryote

13 Suggested Resources for Reading

1. Human Parasitilogy, Bogitsh, B.B., Carter, C.E., Oeltmann, T., Elsevier Science Publishing Co Inc.
2. Clinical Parasitology, A Practical Approach, Zeibig, E.A., Elsevier
3. Veterinary Parasitology, Taylor, M. A., Coop, R. L., Wall, R.L., John Wiley and Sons Ltd,
4. Medical Entomology: A Textbook on Public Health and Veterinary Problems Caused by Arthropods, Eldridge, B.F., Edman, J.)
5. Medical and Veterinary Entomology, Second Edition, Mullen, G.R., Durden, L.A.
6. Medical Entomology for Students, Service
7. Text book of Medical Lab Technology, Godkar, P.B. and Godkar, D.P., Bhalani Publisher

<table>
<tr><td>1</td><td>Course Title</td><td>Bacteriology, Mycology and Virology</td></tr>
<tr><td>2</td><td>CourseCode</td><td>MLMDT 303</td></tr>
<tr><td>3</td><td>Typologyof Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration/ LP (Lectures)
(Practical)</td><td>60 hours of lectures for Theory + 30 hours of practicals
(1 credit = 15 hours of theory learning based on 3 hrs./week)
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h. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
</td></tr>
<tr><td>7</td><td>Job Role</td><td>Medical Lab Technologist.</td></tr>
<tr><td>8</td><td>Course Objectives</td><td>Objective: To give an overview of various aspects of General microbiology and its application</td></tr>
<tr><td>9</td><td>Course Outcomes</td><td>On successful completion of the course, the Students will be able to
CO1. Learn the various classifications of bacteria.
CO2. Understand the pathogenicity, transmission of various Bacteria,.
CO3.Understand the various Viruses, Fungi, their detailed study.</td></tr>
<tr><td>10</td><td>Pedagogy of the Course Work</td><td>Classroom Lecture, Laboratory, Interactivesessions, PPT, Practicaland Group Discussions, Case Studies etc.</td></tr>
<tr><td>11</td><td>Instructions for paper setters and candidates</td><td>5. The syllabus of this paper has been divided into FOUR units.
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8. All questions carry equal marks.</td></tr>
</table>

12 **Course Content**

UNIT– I

- Introduction to important diseases caused by Streptococcus, Pneumococcus, Neisseria, Corynebacterium, Bacillus, Vibrio cholerae, Yersinia, Hemophilus, Mycobacterium. The operative Laboratory identification of infectious agents, Diagnosis of anaerobic infections, identifying.

- Systematic grouping of pathogenic bacteria pathogenic mechanisms, laboratory diagnosis, prevention and control of these diseases.

- Characteristics of common pathogenic bacteria, IMViC, Urease, catalase, gelatin liquefaction, coagulase, oxidase, sugar fermentation.

UNIT II

- Introduction to Human mycotic infections viz. Blastomycosis, Opportunistic Mycosis, Candidiasis and Aspergillosis, Cryptococci, Dermatophytes, Sporotrichoums, Histoplasma, Blastomyces, Dematiaceous.

- Antimicrobial sensitivity testing and assay methods for body fluids. Preparation and standardization fungi, Mycetoma, Nocardia and common laboratory contaminants. Biochemical tests for fungi antigens and antisera

UNIT–III

- Chemotherapy of Viral diseases, Oncogenic Viruses, RNA/DNA Viruses, AIDS, Miscellaneous
- Morphology, pathogenesis, life cycle, laboratory diagnosis, prevention and control of viral diseases
- Structure of viruses, lysogenic cycle, lytic cycle,viruses viz. Rabies, Polio, Small pox, Herpes, Measles, Influenza and AIDS
- Collections, handling and storage of samples for viral diagnosis

UNIT-IV

- Principles of biosafety hoods. Use of pipettes, syringes and other virus contaminated instruments in the laboratory Different staining techniques used in virology
- Demonstration of preservation of viruses, viral antigens, infected biological materials and viruses
- Mode of transmission of viral agents, Prevention of viral diseases
- Use of Embryonated eggs in clinical Virology
- Principles of animal cell culture and their use in virology

Practical:

1. Demonstration of staining procedures: Preparation of the following stains and demonstration of 25 viral inclusion bodiesa) Seller's stain for Negri body demonstration b) Giemsa Stain for CMV and Herpes viral inclusions
2. Preparation of reagents for serological tests: Phosphate buffered saline, Dextrose gelatin and Tris buffer3. Principles and performance of Haemagglutination test
3. Demonstration of complement fixation test
4. Demonstration of Immunofluorescence test and lmmunoperoxidase test
5. Demonstration of ELISA for HbsAg detection
6. 7. Biochemical Confirmatory test for Bacteria

13 **Suggested Resources for Reading**

1. Mims' Medical Microbiology Richard Goering , Hazel Dockrell , Mark Zuckerman , Ivan M. Roitt , Professor Peter L. Chiodini Publisher Elsevier Health Sciences

2. Ananthnaryans and Paniker's Text book of Microbiology,CKJ paniker Orient Longman

3. Roitt's Essential immunology Delves, Peter J., Martin, seamus J.Burton,Dennis R.Roitt, Ivan M

Ananthanarayan and Paniker's Textbookof microbiology Kapil , arti ed
4. Microbiology: an Introduction, Gerard J. Tortora Berdell R. Funke and Christine L. Case
5. Parasitology Chatterjee K.D.
6. Microbiology Pelczar, Michal J and Others
7. Medical microbiology Greenwood David and Other
8. Ananthanarayan and Panikar's text book of Microbiology Arti kapil
9. Immunology Male David and Other
10. Mackie and Mc Cartney practical Medical Microbiology Collee J.G and Other
11. Bailey and Scott's Diagnostic Microbiology, Patricia Tille
12. Sherris Medical Micrbiology Pottinger, L. Barth Reller and Charles R. Sterling

Semester IV

<table>
<tr><td>1</td><td>Course Title</td><td>Clinical Laboratory Management</td></tr>
<tr><td>2</td><td>CourseCode</td><td>Skill MLMDT 401</td></tr>
<tr><td>3</td><td>Typologyof Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration/ LP (Lectures)</td><td>90 hours of lectures for Theory
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<tr><td>5</td><td>Credits</td><td>4 Credits</td></tr>
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l. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
</td></tr>
<tr><td>7</td><td>Job Role</td><td>Medical Lab Technologist.</td></tr>
<tr><td>8</td><td>Course Objectives</td><td>● To Understand the importance and method of Observing and reporting while dealing with patients
● Understand Guidelines for Collecting documentation , To maintain restful environment.</td></tr>
<tr><td>9</td><td>Course Outcomes</td><td>On successful completion of the course, the Students will be able to
CO1.Gain the sufficient knowledge of laboratory and its organization.
CO2.Understanding ,managing , supervising ,reporting in lab
CO3. Learning professional behavior in healthcare setting</td></tr>
<tr><td>10</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
<tr><td>11</td><td>Instructions for paper setters and candidates</td><td>9. The syllabus of this paper has been divided into FOUR units.
10. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
11. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
12. All questions carry equal marks.</td></tr>
</table>

12 Course Content

UNIT– I

- Organization of Laboratory:
- Functional components of clinical laboratories (cleanliness, precautions to be taken w.r.t. patients,
- Principles and processes for providing customer and personal services including needs, assessment
- How to deal with various people
- Basic needs of clinical laboratory technician, awareness of soft skills
- Communication between physician, patients and the medical laboratory professional reports, analysis.
- Quality service standards, alternative delivery systems and customer satisfaction evaluation techniques.

UNIT– II

- Quality control of clinical investigations, Automation in clinical biochemistry laboratory, laboratory
- Observing and reporting: Understanding the importance and method of Observing and reporting while dealing with patients
- Management and maintenance of records
- during sample and report collection Importance and method of Observing and reporting while assisting the pathologists and other
- The principles of leadership and guidance
- Managing people/patient effectively as per the guidelines
- Supervising and guiding other laboratory personnel
- Role and importance of assisting other healthcare providers in delivering care
- The relevant legislation, standards, policies and procedures followed in the Organization
- members of the team Importance of verbally informing the person in authority
- Research procedures and protocol: The process of generating or using different sets of rules to combine or group things in different ways

- Documentation

UNIT– III

- Guidelines for documentation, Guidelines for Collecting documentation Various types of records in laboratory set up
- Uses and importance of records in laboratory set up
- Understand abbreviations and symbol
- Essential requirement of records
- Professional Behavior in Healthcare Settings
- Usage of LIMS (Laboratory information management system)

- Enter, transcribe, record, store, or maintain information in written or electronic/magnetic form

UNIT– IV

- How to maintain restful environment. Business, mission, and objectives of the organization
- General and Specific etiquettes to be observed on duty
- Understand need for compliance of organizational hierarchy and reporting
- Legal and ethical issues
- Importance of conservation of resources in laboratories
- Effective working relationships with people, The importance of planning, prioritizing and organizing work, Efficient use of time, Importance of keeping the work area clean and tidy

Suggested Resources for Reading:

1REFERENCE BOOKS

1. Clinical Laboratory Management, Lynne Shore

2. Medical Laboratory Management, Sangeeta Sharma et al Biostatistics : A Foundation for Analysis in Health Sciences,Wayne W. Daniel Business Communication, Sinha, K. K.

3. Business English, Taylor, Ken.

4. English Vocbulary in Use, Mccarthy, Micheal.

5. Business Letters, Emails, Shirley Taylor

6. The Craft of Business Letter Writing, Monipally, M. Matthew.

7. Presentations Laws, Anne

8. Telephoning and Teleconferencing Skills, Ken Taylor

9. Write to the Point : How to Communicate in business with style and purpose, Salvatore J. Iacone

10. Commercial Correspondence and Office Management, R. S.N. Pillai

11. Comdex Computer Course Kit, Gupta, V.

Computer Fundamentals, P.K. Sinha

<table>
<tr><td>1</td><td>Course Title</td><td>Histopathology, Cytology Techniques and Coagulation</td></tr>
<tr><td>2</td><td>CourseCode</td><td>MLMDT 402</td></tr>
<tr><td>3</td><td>Typologyof Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration/ LP
(Lectures)
(Practical)</td><td>60 hours of lectures for Theory + 30 hours of practicals
(1 credit = 15 hours of theory learning based on 3 hrs./week)
(1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)
3-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>5</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>6</td><td>Max. Marks and Time</td><td>
<table>
<tr><td>External Theory</td><td>Internal Theory</td><td>External Practical</td><td>Internal Practical</td></tr>
<tr><td>40</td><td>10</td><td>40</td><td>10</td></tr>
</table>
100Marks (80TH/PR+20IA)
i. The exam of this course will be of 3hours duration.
j. There will be no objective type questions.
k. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
l. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
</td></tr>
<tr><td>7</td><td>Job Role</td><td>Medical Lab Technologist.</td></tr>
<tr><td>8</td><td>Course Objectives</td><td>: To apprise students of the basics of histopathology and Cytology techniques. Further, understand the relevance of blood coagulation in diagnosis.</td></tr>
<tr><td>9</td><td>Course Outcomes</td><td>On successful completion of the course, the Students will be able to
CO1. To gain broad understanding of histology ,cytology .
CO2.To understand the various staining techniques .
CO3. To study microtomy.</td></tr>
<tr><td>10</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
<tr><td>11</td><td>Instructions for paper setters and candidates</td><td>9. The syllabus of this paper has been divided into FOUR units.
10. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
11. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
12. All questions carry equal marks.</td></tr>
</table>

12 Course Content

UNIT– I

- General understanding of the terms –Histology, Cytology, Histopathology and Histopathological techniques. General organization of histopathological laboratory and basic requirements of histopathology laboratory Responsibilities of a histotechnologist
- Basic steps for Tissue Processing- Fixation, Embedding, Microtomy, Staining, Mounting, methods o of decalcifications .
- Chemicals and Reagents, Equipments- Microscope, Microtome- Types, Uses, Parts, different types microtome knives, care and maintenance. Automated tissue processor- Components, working and precautions during use. Tissue floating bath
- Updates on advanced techniques and future trends in field of tele-pathology

UNIT– II

- Methods of Cytological Staining
- Staining Methods: Hematoxylin and Eosin stain
- PAP staining: Components and methods
- Hematoxylin: Types, methods of preparation, staining, Eosin- Method of preparation. Reticulin stain

UNIT-III

- Mechanism of blood coagulation- Extrinsic Pathway, Intrinsic Pathway. Regulators of blood coagulation
- Testing of blood coagulation: Bleeding Time, Duke's method
- Clotting Time: Capillary tube method and Lee white's method, PT, aPTT, TT, Clot retraction time, Determination of fibrinogen

- Sample collection technique (Phelbotomy)

UNIT IV

- Platelet disorder: Thrombocytopenias - causes including aplastic anemia, Hemophilia
- Laboratory diagnosis and investigation of Bleeding disorders
- Anticoagulant and diseases: Sample preparation, Anticoagulant used in testing and treatment

Practical:

1. Precautions to prevent hemolysis
2. Storage of blood specimens
3. Bleeding time & clotting time estimation
4. Prothrombin time estimation
5. aPTT (activated partial thromboplastin time) estimation
6. Clot retraction time
7. Parts of microtome
8. UNIT Cutting of the tissue and staining
9. Tissue processing
10. 10 PAP staining

13 Referencebooks

1 Gartner, L.P., Text Book of History, Elsevier

2. Steven, A., Lome, J.S., & Yerng, B., Wheater's Basic Histology, Churchill Livingstone

3. Stacey e. Mills, Histology for pathologist

4. Godkar, P.B., Text book of medical laboratory technology Bhalani

1	**Course Title**	**Endocrinology, Tumor and Cancer Markers**
2	**CourseCode**	**MLMDT 403**
3	**Typologyof Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**60 hours of lectures for Theory + 30 hours of practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **3-groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	

External Theory	Internal Theory	External Practical	Internal Practical
40	10	40	10

100Marks (80TH/PR+20IA)

i. The exam of this course will be of 3hours duration.
j. There will be no objective type questions.
k. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
l. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees

7	**Job Role**	**Medical Lab Technologist.**
8	**Course Objectives**	. **Objective: : To gain knowledge about endocrinology, tumor and cancer markers**
9	**Course Outcomes**	**On successful completion of the course, the Students will be able to** **CO1.** Learn the various hormones and enzymes. **CO2.** Understand the physiology and chemistry of hormones **CO3.**Understand the tumor markers in detail.
10	**Pedagogy of the Course Work**	ClassRoomLecture, Laboratory,Interactivesessions,PPT,Practicaland GroupDiscussions,Case Studiesetc.
11	**Instructions for paper setters and candidates**	9. **The syllabus of this paper has been divided into FOUR units.** 10. **Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** 11. **The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.** 12. **All questions carry equal marks.**

Course Content

UNIT– I

- Introduction to hormones, difference between hormones and enzymes, classification of hormones.
- Regulation and general mechanism of action of hormones, Pituitary gland and hypothalamus
- Hormones of the Anterior Pituitary: Growth hormone, Prolactin, Gonadotropin, Follicle Stimulating hormone, Luteinizing Hormone, Adrenocorticotropic hormone (ACTH).
- Hormones of neurohypophysis: Oxytocin, Antidiuretic hormone (ADH).

UNIT II

- Hormones of the Thyroid gland: Chemistry and normal physiology, Thyroid disorders- Goiter, myxodema, autoimmune thyroiditis, tumors of the thyroid gland, hyperthyroidism, Graves' disease.
- Hormones of the gonads: Testosterone, Estrogens, Progesterone, their synthesis and functions
- Cortisol and functions, Cushing's syndrome, Conn's syndrome
- Human Chorionic Gonadotropin (HCG) hormone, menstrual cycle, Menopause. Hormone of pancreasInsulin- Its metabolic effects on carbohydrates, fats and protein, control of insulin secretion. Blood glucose regulation, Diabetes Mellitus, Somatostatin
- Hormone of kidney: Renin

UNIT III

- Introduction, The Carcinogens- Definition, Oncogene- Definition- Mechanism of action of Oncogenes Characteristics of growing tumor cells- General and morphological biochemical changes.
- Tumor Markers- Introduction and definition, Clinical applications of tumor markers
- Enzymes as tumor markers, Alkaline Phosphatase (ALP), Creatine kinase (CK), Lactate dehydrogenase (LDH), Prostatic acid phosphatase (PAP), Prostate specific antigens (PSA)

UNIT-IV

- Hormones as tumor markers (introduction of each type in brief): Oncofetal antigens, Alpha feto protein (AFP), Carcino embryonic antigen (CEA)
- Carbohydrate markers (brief introduction of each type): CA 15-3, CA 125
- Bladder cancer markers (introduction in brief): Bladder tumor antigen (BTA)
- Blood group antigen (brief introduction of each type): CA 19-9, CA 50, CA 72-4, CA 242

Practical:

1. Blood grouping and Cross Matching
2. ELISA
3. Estimation of T3
4. Estimation of T4
5. Estimation of TSH
6. Estimation of FSH
7. Estimation of LH
8. Estimation of hCG
9. Estimation of Progesterone
10. Estimation of Testosterone
11. Estimation of CA- 125
12. Test for HIV
13. Test for Hepatitis B (HBsAg)
14. Test for Hepatitis (HCV)
15. Malaria antigen
16. Tuberculosis-lgG/Ig

13

Reference Books:

1. Levin, N., Manual of Endocrinology and Metabolism
2. Headley, M.E., & Levine, J.E., Endocrinology, Pearson Education
3. Rochelle, W.J. and Shimkets, R.A., Ongenomics Handbook, Humana Presss
Wang, E., Cancer Systems Biology, CRC press

1	**Course Title**	**Microbiology -I**
2	**CourseCode**	**404**
3	**Typologyof Course**	Core Discipline
4	**Course Duration/ LP** (Lectures) (Practical)	**60 hours of lectures for Theory + 30 hours of practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **3-groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	100Marks (80TH/PR+20IA) **m.** The exam of this course will be of 3hours duration. **n.** There will be no objective type questions. **o.** Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. **p.** Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees

External Theory	Internal Theory	External Practical	Internal Practical
40	10	40	10

7	**Job Role**	**Medical Lab Technologist.**
8	**Course Objectives**	**To learn the techniques of collection of samples, their processing and the identifications of the various pathogens, like bacteria, parasites, viruses, using different techniques.** **To provide basic knowledge of the different diseases caused by various microorganisms is also imparted, their processing and the identifications of the various pathogens, like bacteria, parasites, viruses, using different techniques.**
9	**Course Outcomes**	**On successful completion of the course, the Students will be able to** **CO1. To provide basic knowledge of the different diseases caused by various microorganisms.** **CO2. To provide training in the use of standard safety measures while handing highly infected material**
10	**Pedagogy of the Course Work**	ClassRoomLecture, Laboratory,Interactivesessions,PPT,Practicaland GroupDiscussions,Case Studiesetc.
11	**Instructions for paper setters and candidates**	**1.** The syllabus of this paper has been divided into FOUR units. **2.** Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. **3.** The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. **4.** All questions carry equal marks.

12

Course Content UNIT– I

- Elementary knowledge of Pyogenic cocci, Gram Negative Bacilli, Gram Positive bacilli, anaerobes, mycobacteria and spirochates
- Difference between cocci and rods
- Normal flora of human body

UNIT II

- Lab diagnosis of common Bacterial infections viz. pyogenic infections, Respiratory tract infections, Meningitis, Diphtheria, Whooping Cough, Gas gangrene, food poisoning, Acute diarrhea diseases, cholera, Urinary tract infection, Tuberculosis, Plague, Anthrax, Typhus fever, syphilis, Gonorrhea and other STD's

UNIT III

- Disease oriented microbiology, culture & sensitivity test, , aerobic and anaerobic techniques.
- Introduction to parasitic fungi, specimen collection, Laboratory diagnosis of mycotic infections, Diagnostic mycology
- Principles and mode of action of antibiotics and chemotherapeutic agents for bacteria and fungi
- Principles of Antigen-Antibody reactions
- Diagnostic Microbiology & Micro Technique

UNIT-IV

- Role of microbiology laboratory, Quality control in microbiology, Quick reference of media and biochemical tests
- Collection and handling of fecal specimen, Laboratory techniques in parasitological investigation of stool, Processing of specimens other than stool, Lab identification of human parasites
- Collection, transportation and processing of clinical samples for microbiology investigations
-

Practical:

1. Preparation of Smear
2. Bacteriophage typing methods
3. Monochrome staining (simple staining)
4. Gram's staining
5. Study of motility of bacteria: Hanging Drop Method
6. Isolation of bacteria by streak plate techniques
7. To perform qualitative WIDAL test
8. Bacterial Growth Curve using Spectrophotometer
9. To check the effect of temperature on bacterial grow

13

Reference Books:

1. Mims' Medical Microbiology Richard Goering, Hazel Dockrell, Mark Zuckerman, Ivan M. Roitt, Professor Peter L. Chiodini, Elsevier Health Sciences

2. Roitt's Essential Immunology Delves, Peter J., Martin, Seamus J. Burton, Dennis R. Roitt,

Ivan M Ananthanarayan and Paniker's Textbook of Microbiology Kapil
3. Microbiology: An Introduction, Gerard J. Tortora Berdell R. Funke and Christine L. Case
4. Parasitology Chatterjee K.D.
5. Microbiology Pelczar, Michal J and Others
6. Medical Microbiology Greenwood David and Other
Ananthanarayan and Panikar's Text Book of Microbiology Arti kapil

Semester V

1. Course Title Clinical Biochemistry

2. **Course Code** MLMDT 501
3. **Typology of Course** Core discipline
4. **Course Duration/LP (lectures)** 90 hours lectures of theory (1 credit = 15 hours of theory learning on 6 Hrs/week

5. **Credits** 4 credits
6. **Max. Marks and Time**

External Theory	Internal Theory	External Practical	Internal Practical
40	10	40	10

100Marks (80TH/PR+20IA)

a. The exam of this course will be of 3hours duration.
b. There will be no objective type questions.
c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.

7. **Job role** Medical lab technologist
8. **Course Objective** On successful completion of the course, the Students will be able to :
   - Understand basic biochemical processes.
   - Learn common clinical laboratory tests.
   - Understand the role of biochemistry in disease diagnosis.
9. **Course Outcomes**
   - Explain major metabolic pathways.
   - Perform basic biochemical tests.
   - Interpret common clinical laboratory results.
10. **Pedagogy of the Course work**: Class room lecture, laboratory, interactive sessions. PPT, Practical and group discussion

11. **Instructions for Paper Setter and candidates**:

    - The syllabus of this paper has been divided in to FOUR units.
    - Examiner will set a total of NINE questions comprising two questions from each unit, including Question no. 1 (compulsory) of short answer type covering the whole syllabus.
    - The students are required to attempt one question from each unit and entire Compulsory Question no. 1
    - All questions carry equal marks.

Syllabus

SEMESTER V

MLMDT 501 Clinical Biochemistry

UNIT I

- Clinical enzymology: Diagnostic enzymes-enzyme assay in serum/plasma, urine and cells. Clinically important enzymes, use of isoenzymes in diagnosis
- Organ function tests: Assessment and clinical manifestation of hepatic, renal, gastrointestinal and pancreatic functions
- Enzyme Histochemistry and demonstrations of phosphatases, dehydrogenases oxidases and peroxidases etc.

UNIT II

- General reactions of amino acids: transamination, role of SGOT and SGPT and their clinical significance
- Determination of serum glutamate pyruvate transaminase (SGPT) and serum glutamate Oxaloacetate transaminase (SGOT) End point reaction
- Glucose tolerance test, insulin tolerance test, gastric analysis, xylose absorption test, Creatine Phosphokinase (CPK), CPK-MB

UNIT III

- Liver function tests: Estimation of important Amino acids and enzymes- Enzymes Acid Phosphatase (ACP), Alkaline Phosphatase (ALP), Transaminase (ALT), LDH, Amino transferase (AST), Amylase, albumin, bilirubin

UNIT IV

- Renal function tests: Renal calculi, Creatinine, clearance test for renal function.
- Endocrine function tests
- Lipid profile: HDL, LDL, Serum cholesterol

Practicals:

1. Liver tests, Renal tests, Endocrine function tests, Lipid profile,
2. Estimation of important Amino acids and enzymes: Enzymes Acid Phosphatase (ACP), Alkaline Phosphatase (ALP), Transaminase, LDH, AST, ALT, Amylase and lactate dehydrogenase, Creatinine, Creatine Phosphokinase (CPK) , CPK-MB.
3. Determination of serum glutamate pyruvate transaminase (SGPT) and serum glutamate Oxaloacetate transaminase (SGOT) End point reaction

Reference Books

1. A guidebook to Biochemistry, Yudkin, M.
2. A Manual of Laboratory & Diagnostic Tests, Frances Fischbach
3. Biochemistry, Voet and Voet
4. Biochemistry, Stryer
5. Biochemistry, Satyanarayan., S. & U. Chakrapani
6. Clinical Biochemistry, Richard Luxton
7. Clinical Diagnosis & Management by Laboratory methods, Henary, J.B.
8. Clinical Biochemistry, Guru, G.
9. Handbook of Biochemistry Siddique, M.A.
10. Textbook of Medical Biochemistry, Ramkrishnan, S.
11. Biochemical Techniques, K. Choudhary
12. Text book of Medical Biochemistry, Chaterjee&Shinde
13. Principles of Biochemistry, David L. Nelson
14. Principles of Biochemistry, Lehninger
15. Textbook of Biochemistry and Human Biology, G.P. Talwar
16. Textbook of Medical Laboratory Technology, Godkar and Godkar
17. Outline of Biochemistry, Conn Stumpf
18. Principles of Internal Medicine, Isselbacher
19. Proteins and Proteomics : Laboratory Manual, Richard J. Simpson
20. Purifying Proteins for Proteomics: Laboratory Manual, Richard J. Simpson
21. Enzymes: Biochemistry, Biotechnology & Clinical chemistry,Palmer Trevor, Publisher: Horwood Pub. Co.
22. Outlines of Biochemistry, Erice Conn & Paul Stumpf ; John Wiley and Sons
23. Fundamentals of Biochemistry. Donald Voet& Judith Voet , John Wiley and Sons, Inc.
24. Lehninger, Principles of Biochemistry, David Nelson & Michael Cox, W.H. Freeman and Company.
25. Biochemistry, Jeremy Berg, LubertStryer, W.H.Freeman and Company
26. Biochemical Methods for Agricultural Sciences, Sadasivam and Manikam. Wiley Eastern Limited
27. Practical Clinical Biochemistry, Harold Varley, CBS

Semester V

1. Course Title Basics of Molecular Biology

2. **Course Code** MLMDT 502
3. **Typology of Course** Core discipline
4. **Course Duration/LP (lectures)** 90 hours lectures of theory (1 credit = 15 hours of theory learning on 6 Hrs/week

5. **Credits** 4 credits
6. **Max. Marks and Time**

External Theory	Internal Theory	External Practical	Internal Practical
40	10	40	10

100Marks (80TH/PR+20IA)

a. The exam of this course will be of 3hours duration.
b. There will be no objective type questions.
c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.

7. **Job role** Medical lab technologist
8. **Course Objective** On successful completion of the course, the Students will be able to :

- To understand the structure and function of DNA andRNA.
- To study DNA replication, transcription and translation.
- To understand gene expression and regulation.

9. **Course Outcomes**
   - Explain the structure and functions of DNA and RNA.
   - Describe replication, transcription and translation.
   - Explain gene regulation and mutations.

.

10. **Pedagogy of the Course work**: Class room lecture, laboratory, interactive sessions. PPT, Practical and group discussion

11. **Instructions for Paper Setter and candidates**:

   - The syllabus of this paper has been divided in to FOUR units.

   - Examiner will set a total of NINE questions comprising two questions from each unit, including Question no. 1 (compulsory) of short answer type covering the whole syllabus.
   - The students are required to attempt one question from each unit and entire Compulsory Question no. 1
   - All questions carry equal marks.

Syllabus

MLMDT 502 Basics of Molecular Biology

UNIT I

- Genetics: Mendel Law of Inheritance.
- Chromatin structure, DNA binding protein – Histones and non- histone proteins, structure , Supercoiling of DNA, positive and negative supercoiling
- Replication of DNA: Concept of replication, uni- and bi- directional Replication- Initiation, elongation and termination of replication, Enzymes and accessory proteins, Telomerases- Mechanism and role in cancer

UNIT II

- Transcription: Mechanism of transcription in prokaryotes and eukaryotes
  - Synthesis of rRNA and mRNA, transcription factors

- Post Transcriptional Modifications
- Translation: Process, Translation machinery, Ribosomes, ER, Composition and assembly.

UNIT III

- Gene regulation: Transcription regulation in prokaryotes- Principles of transcriptional regulation with examples from Lactose and Tryptophan operons.
- Transcription regulation in eukaryotes- Activators, repressors, enhancers, silencer elements, TATA box, CAT box

UNIT IV

- Genetic code, Biochemical elucidation of codon and its properties.
- Post-translational modifications.
- Mutation: Introduction and Types of Mutation, Introduction, Types and Properties.
- DNA Damage, Repair and Recombination. Transposable elements

Practical:

1. Isolation of Genomic DNA from Blood
2. Isolation of Plasmid from Bacteria
3. Agarose Gel Electrophoresis

Reference books:

1 . Lewin, B., Gene VIII and Gene IX, Pearson
2 . Albert, B., Johnson, A., Raff, M., Roberts, K. & Walter, P., Molecular Biology of the Cell. Garland Science
3 . Karp, G., Iwan, J., Marshall, W., Cell and molecular Biology, Concepts and Experiments, Wiley
4 . Nelson LD and Cox, M., Lehninger Principles of Biochemistry, W.H. Freeman
5 . Berk, A., Kaiser, C.A., Lodish, H., Amon, A., Ploegh, H., Bretscher, A., Kreiger, M., &Martnin, K.C.,
6 Molecular Biology of the Cell, W.H. Freeman
7 . Primerose, S.B., Tonyman, R., & Old, B., Principle of Gene Manipulation : An Introduction to Genetic Engineering, Wiley-Blackwell.
8 Brown, T.A., Gene Cloning : An Introduction, Nelson Thomes Ltd.
9 Watson, J.D., Candy, A.A, Myers, R.H. &Witkowski, J.A., Recombinant DNA, WH Freeman

Semester V

1. Course Title Immunology and Serology

2. Course Code MLMDT 503
3. Typology of Course Core discipline
4. Course Duration/LP (lectures) 90 hours lectures of theory (1 credit = 15 hours of theory learning on 6 Hrs/week

5. Credits 4 credits
6. Max. Marks and Time

External Theory	Internal Theory	External Practical	Internal Practical
40	10	40	10

100Marks (80TH/PR+20IA)

a. The exam of this course will be of 3hours duration.
b. There will be no objective type questions.
c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.

7. Job role Medical lab technologist

8. Course Objective On successful completion of the course, the Students will be able to :

- •To understand the basic concepts of immunity
- To study antigens, antibodies and their interactions.
- To understand humoral and cell-mediated immunity.

9. Course Outcomes

- Explain the basic principles of the immune system.
- Describe the structure and functions of antigens and antibodies.
- Differentiate between innate and adaptive immunity

10. Pedagogy of the Course work: Class room lecture, laboratory, interactive sessions. PPT, Practical and group discussion

11. Instructions for Paper Setter and candidates:

- The syllabus of this paper has been divided in to FOUR units.
- Examiner will set a total of NINE questions comprising two questions from each unit, including Question no. 1 (compulsory) of short answer type covering the whole syllabus.
- The students are required to attempt one question from each unit and entire Compulsory Question no. 1
- All questions carry equal marks.

Syllabus

MLMDT 503 Immunology and Serology

UNIT I

- Introduction to immune system: Innate and adaptive immunity, Cells and tissues of immune system, Functions of lymphoid tissue, Properties of immunogen, Hapten, adjuvants, epitopes
- Antibody and MHC: Basic structure of antibody, Major classes and their biological activity, Antigenic determinants, Structure, function relationships in antibody. Major histocompatibility complex (MHC), Structure and properties of class I and II MHC, Expression of MHC molecule, Overview of monoclonal antibody

UNIT II

- Cell mediated immune response, T-cell activation and differentiation, Cytotoxic T cells and its functions, Humoral response, B-cell activation and differentiation, Complement system
- Immune system in health and disease: Dysfunctional immunity, Hypersensitivity reactions- Type I to Type IV reactions, Immunodeficiency diseases, Autoimmune diseases, Transplantation immunology

UNIT III

- Vaccine: Active and passive immunization, Designing of vaccine for active immunization, Live attenuated vaccine, Inactivated vaccine, DNA vaccine, Recombinant vector vaccine, Cancer and immune system
- Immune response to tumors, Immunotherapy

UNIT IV

- Serological reactions: Antigen antibody reactions, Precipitation reactions, Agglutination reactions, Radioimmunoassay and ELISA, Western blotting reactions, Immunofluorescence Flow cytometry

Practical:

1. To perform indirect method of blood grouping
2. To Separate Serum from blood
3. To Separate Plasma from blood
4. CRP Test
5. Immunodiffusion
6. To perform C-reactive protein test (CRP)
7. ELISA test

Reference Books:

1. W., Stanford Serology and immunology : A Clinical Approach, MacMillan
2. Kerby, J., Immunology, WH Freeman
3. Abbar, A., Cellular and Molecular, Saunders
4. Stites, D., Basic and Clinical Immunology, Lange

Semester V

1. Course Title Basics of rDNA technology

2. **Course Code** MLMDT 504
3. **Typology of Course** Core discipline
4. **Course Duration/LP (lectures)** 90 hours lectures of theory (1 credit = 15 hours of theory learning on 6 Hrs/week

5. **Credits** 4 credits
6. **Max. Marks and Time**

External Theory	Internal Theory	External Practical	Internal Practical
40	10	40	10

100Marks (80TH/PR+20IA)

a. The exam of this course will be of 3hours duration.
b. There will be no objective type questions.
c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.

7. **Job role** Medical lab technologist
8. **Course Objective** On successful completion of the course, the Students will be able to :
   - To understand the basic principles of recombinant DNA technology.
   - To study restriction enzymes, vectors and DNA ligase.
   - To understand the steps involved in gene cloning
9. **Course Outcomes**

- Explain the principles of rDNA technology.
- Identify different enzymes and vectrs used in gene cloning.
- Describe the basic process of recombinant DNA formation and cloning.

10. **Pedagogy of the Course work**: Class room lecture, laboratory, interactive sessions. PPT, Practical and group discussion

11. **Instructions for Paper Setter and candidates**:

- The syllabus of this paper has been divided in to FOUR units.
- Examiner will set a total of NINE questions comprising two questions from each unit, including Question no. 1 (compulsory) of short answer type covering the whole syllabus.
- The students are required to attempt one question from each unit and entire Compulsory Question no. 1
- All questions carry equal marks.

Syllabus

MLMDT -504 Basics of recombinant technology

UNIT I

- Application of rDNA technology ,History and milestones in genetic engineering
- What is recombinant DNA, tools of recombinant DNA technology , enzymes , markers , probes, ladders

UNIT II

- Vectors, plasmids,cosmids, phages, viral vectors,expression vectors.
- Gene cloning techniques:DNA isolation, restriction digestion , ligation, transformation

UNIT III

- Selection and screening of recombinants,Expression of recombinant proteins: host systems, expression control.

UNIT IV

- Polymerase chain reaction, principle , steps, primers designing, types of PCR, Applications of pcr.

Practicals

1 DNA isolation from bacteria
2 Restriction digestion of plasmid DNA
3 Ligation
4 Bacterial transformation
5 Screening
6 PCR

Semester V

1. Course Title Introduction to Biochemical Techniques

2. **Course Code** MLMDT 505
3. **Typology of Course** Core discipline
4. **Course Duration/LP (lectures)** 90 hours lectures of theory (1 credit = 15 hours of theory learning on 6 Hrs/week

5. **Credits** 4 credits
6. **Max. Marks and Time**

External Theory	Internal Theory	External Practical	Internal Practical
40	10	40	10

100Marks (80TH/PR+20IA)

a. The exam of this course will be of 3hours duration.
b. There will be no objective type questions.
c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.

7. **Job role** Medical lab technologist
8. **Course Objective** On successful completion of the course, the Students will be able to :
   - To understand the basic principles of biochemical techniques.
   - To learn common laboratory instruments and procedures.
   - To understand methods used for qualitative and quantitative analysis.

9. **Course Outcomes**
   - Explain the principles of basic biochemical techniques.
   - Operate common biochemical laboratory instruments.
   - Perform basic qualitative and quantitative biochemical tests.

10. **Pedagogy of the Course work**: Class room lecture, laboratory, interactive sessions. PPT, Practical and group discussion

11. **Instructions for Paper Setter and candidates**:

- The syllabus of this paper has been divided in to FOUR units.
- Examiner will set a total of NINE questions comprising two questions from each unit, including Question no. 1 (compulsory) of short answer type covering the whole syllabus.
- The students are required to attempt one question from each unit and entire Compulsory Question no. 1
- All questions carry equal marks.

Syllabus

MLMDT 505 Introduction to biochemical Techniques

UNIT I

Spectroscopic techniques:

- Beer-Lambert's Law. Light absorption and its transmittance. Determination and application of extinction coefficient
- Applications of following spectroscopic techniques in elucidating structure of Biomolecules: Visible, U.V., infra-red and fluorescence spectroscopy. ORD, C.D. and N.M.R.

UNIT II

Electrophoretic Techniques:

- Principles and applications of the following electrophoresis techniques: Paper and gel electrophoresis, high voltage electrophoresis, SDS-PAGE, Discontinuous electrophoresis, isotachophoresis, isoelectric focusing and immunoelectrophoresis
- Centrifugation Techniques: Various centrifugation techniques and their applications in Biochemistry. Preparative and analytical ultra-centrifugation procedures. Application of partial specific volume, diffusion coefficient and viscosity measurements in the study of macromolecules of biochemical importance

UNIT III

Chromatographic Techniques:

- Introduction to Chromatography
- General principles of chromatography and the application of following chromatographic procedures in isolation and purification of biomolecules: Absorption, partition, paper and thin layer chromatography
- Gas liquid chromatography, High performance liquid chromatography (HPLC), Ion exchange and Exclusion chromatography, Affinity chromatography

UNIT IV

Radio Isotopic Techniques:

- Nature of isotopes and radioisotopes. Radioactive decay
- Properties of radioactive emissions, Units of radioactivity, Techniques used to measure radioactivity: GM counter, liquid scintillation counting and gamma counter.
- Biological hazards of radiation and safety measures in handling radioisotopes

Practicals:

1. Separation and identification of amino acids by
   a. Paper chromatography
   b. Thin layer chromatography
   c. Separation of phospholipids by thin layer chromatography
2. Preparation of starch from potato and its hydrolysis by salivary amylase

3. Estimation of lactic acid in blood before and after exercise
4. Gel Electrophoresis of Nucleic Acids

Reference Books

1. Principles and Techniques of Biochemistry and Molecular Biology, Keith Wilson
2. Introduction to Instrumentation in Life Sciences, Prakash Singh Bisen, Anjana Sharma
3. Chemical Analysis Modern Instrumentation Methods and Techniques Second Edition Francis Rouessac and AnnickRouessac
4. Principles and Techniques of Biochemistry and Molecular Biology, Wilson K.M., Walker J.M., Cambridge University Press, UK
5. Biochemical spectroscopy. Vol 46 of Methods in Enzymology.,Kenneth Sauer. Academic Press
6. Modern experimental biochemistry, Rodney Boyer. Prentice Hall
7. Analytical Biochemistry,David Holmes, H.Peck , Prentice Hall
8. An Introduction to Practical Biochemistry,David T Plummer. Tata McGraw-Hill Publishing Company Limited
9. Introductory Practical Biochemistry, S. K. Sawhney, R. Singh

Semester V

1. Course Title Safe laboratory Practices

2. Course Code MLMDT 506
3. Typology of Course Core discipline
4. Course Duration/LP (lectures) 90 hours lectures of theory (1 credit = 15 hours of theory learning on 6 Hrs/week
5. Credits 4 credits
6. Max. Marks and Time

External Theory	Internal Theory	External Practical	Internal Practical
80	20	---	---

100Marks (80TH/PR+20IA)

a. The exam of this course will be of 3hours duration.
b. There will be no objective type questions.
c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.

7. Job role Medical lab technologist
8. Course Objective On successful completion of the course, the Students will be able to :

- To understand basic laboratory safety rules and precautions.
- To learn safe handling of laboratory equipment, chemicals, and biological samples.
- To develop awareness about waste disposal, PPE, and emergency procedures.

9. Course Outcomes

- Students will safely handle specimens, chemicals, and laboratory instruments.
- Students will be able to manage laboratory waste and respond appropriately to emergencies.

10. Pedagogy of the Course work: Class room lecture, laboratory, interactive sessions. PPT, Practical and group discussion

11. Instructions for Paper Setter and candidates:

- The syllabus of this paper has been divided in to FOUR units.
- Examiner will set a total of NINE questions comprising two questions from each unit, including Question no. 1 (compulsory) of short answer type covering the whole syllabus.
- The students are required to attempt one question from each unit and entire Compulsory Question no. 1
- All questions carry equal marks.

Syllabus

MLMDT 506 - Safe Laboratory Practices

UNIT I

Safety & First Aid

Human health and Homeostasis, medical care in India, Medical Laboratories of developing countries .Basic causes of accidents, common types of laboratory accidents Ethics, responsibility, safety measure, Basics of first aid precautions to ensure self-safety

Sample preservation and precautions while Transporting

First aid in laboratory understanding and precautions to ensure Patient's Safety

Common emergency conditions and what to do in medical emergencies . Basics of first aid

UNIT II

Biomedical Waste Management-I. Biomedical waste management in a clinical laboratory

Disposal of used samples, reagents and other biomedical waste: Importance of Biomedical Waste. Categories of bio-medical waste . Disposal of biological samples material. Bio-medical waste – color coding, types of containers, transportation of waste Disposal of Laboratory waste, Means of bio-medical waste treatment, NABL and SOP

Categorize waste according to national, local and organizational guidelines Appropriate approved disposal routes for waste .Appropriate containment or dismantling requirements for waste . How to make the waste safe for disposal

UNIT III

Biomedical Waste Management-II. Organizational and national waste management principles and procedures . Hazards and risks associated with the disposal and the importance of risk assessments and how to provide these .Personal protective equipment required to manage the different types of waste generated by different work activities .Actions and reporting procedures for any accidents, spillages and contamination involving waste External agencies involved in the transport and receipt of your waste . Importance of segregating different types of waste and how to do this .Safe methods of storage and maintaining security of waste and the permitted accumulation times . Current national legislation, guidelines, local policies and protocols which affect work practice.

UNIT IV

Best Practices in Lab

- Sensitization on current best practices in laboratory
- Elementary knowledge on Good Clinical Laboratory Practices (GCLP) of WHO
- Elementary Knowledge of laboratory safety guidance of OSHA (Occupational Safety and
- Health Administration), U.S. Department of Labor
- Elementary Knowledge of other current practices in laboratories used worldwide.

Reference Books

1. Kanai L. Mukherjee &Swarajit Ghosh (Eds) Medical Laboratory Technology, Procedure Manual for Routine Diagnostic Tests
2. Mukherjee, McGraw-Hill Education Med Lab. Technology
3. F. J. Baker, R. E. Silverton : Introduction to Medical Laboratory Technology
4. Tao Le and VikasBhushan First Aid for the USMLE Step 1
5. Rajeev Sharma : First Aid Guide
6. American Red Cross (Author), Kathleen A. Handal - The American Red Cross First Aid and Safety handbook Paperback – Import
7. Dan Wolfe – Smashwords, First Aid and Beyond
8. Ronald A. Bergman - Anatomy Atlases, Anatomy of First Aid: A Case Study Approach

Semester VI

1.Course Title **Advanced techniques in hematology ,cytology&histopathology**

2.Course Code MLMDT 601

3.Typology of Course Core discipline

4.Course Duration/LP (lectures) 90 hours lectures of theory (1 credit = 15 hours of theory learning on 6 Hrs/week

5.Credits 4 credits

6.Max. Marks and Time

External Theory	Internal Theory	External Practical	Internal Practical
40	10	40	10

100Marks (80TH/PR+20IA)

a. The exam of this course will be of 3hours duration.

b. There will be no objective type questions.

c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.

d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.

7.Job role Medical lab technologist

8.Course Objective On successful completion of the course, the Students will be able to :

- To learn advanced diagnostic technique
- To understand modern staining methods
- To develop skills in sample processing

9.Course Outcomes

- To perform advanced laboratory techniques
- To identify abnormal cells and tissues .
- To apply techniques in clinical diagnosis

10.Pedagogy of the Course work: Class room lecture, laboratory, interactive sessions. PPT, Practical and group discussion

11.Instructions for Paper Setter and candidates:

- The syllabus of this paper has been divided in to FOUR units.
- Examiner will set a total of NINE questions comprising two questions from each unit, including Question no. 1 (compulsory) of short answer type covering the whole syllabus.
- The students are required to attempt one question from each unit and entire Compulsory Question no. 1
- All questions carry equal marks.

SEMESTER VI

Syllabus

MLMDT 601 Advanced Techniques in Haematology, Cytology & Histopathology

UNIT I

- Laboratory test for assessing bleeding disorders
- Laboratory investigation for disseminated intravascular coagulation (DIC)

UNIT II

- Mechanism of fibrinolysis test for fibrinolysis
- Platelet function tests and their interpretation

UNIT III

- Techniques available for cytogenetic studies
- Use of Radioisotopes in hematology
- Safety measures for handling Radioisotopes

UNIT IV

- Advanced techniques and future trends in field of hematology& blood banking
- Advanced techniques and future trends in field of clinical pathology
- Advanced techniques and future trends in field of histopathology & cytopathology

Practicals:

1. Laboratory test for assessing bleeding disorders
2. Laboratory investigation for disseminated intravascular coagulation (DIC)
3. Mechanism of fibrinolysis test for fibrinolysis
4. Platelet function tests and their interpretation

Reference Books:

1. Clinical Haematology, Christopher A. Ludlam
2. Practical Haematology, J.B. Dacie
3. Practical Haematology, Sir John
4. Haematology, Emmanuel C.Besa
5. Haematology (Pathophysiological basis for clinical practice), Stephen M. Robinson
6. Haematology for students Practitioners, RamnikSood
7. Hand book of Medical Laboratory Technology, V.H. Talib
8. Atlas of Haematology, G.A. McDonald

Semester VI

1.Course Title :- Tumour and cancer marker

2.Course Co

3.Typology of Course Core discipline

4.Course Duration/LP (lectures) 90 hours lectures of theory (1 credit = 15 hours of theory learning on 6 Hrs/week

5.Credits 4 credits

6.Max. Marks and Time

External Theory	Internal Theory	External Practical	Internal Practical
40	10	40	10

100Marks (80TH/PR+20IA)

a. The exam of this course will be of 3hours duration.

b. There will be no objective type questions.

c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.

d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.

7.Job role Medical lab technologist

8.Course Objective On successful completion of the course, the Students will be able to :

- To understand basic of tumour and cancer markers
- To learn major tumour marker and their clinical significance .
- To learn methods for detection of cancers.

9.Course Outcomes

- To identify important tumour marker
- To interpret tumour marker result.
- To understand their role in cancer diagnosis and monitoring

10.Pedagogy of the Course work: Class room lecture, laboratory, interactive sessions. PPT, Practical and group discussion

11.Instructions for Paper Setter and candidates:

- The syllabus of this paper has been divided in to FOUR units.
- Examiner will set a total of NINE questions comprising two questions from each unit, including Question no. 1 (compulsory) of short answer type covering the whole syllabus.
- The students are required to attempt one question from each unit and entire Compulsory Question no. 1
- All questions carry equal marks.

Syllabus

MLMDT 602 Tumour and Cancer marker

UNIT I

Introduction to Cancer Biology:Concepts of neoplasia: benign vs malignant,Hallmarks of cancer,Genetic and epigenetic changes in tumors,Overview of TumorMarkers,Definition and characteristics of ideal tumormarkers,Diagnostic, prognostic, and predictive markers,Limitations and false positives/negatives

UNIT II

Biochemical Nature of TumorMarkers,Enzyme markers (e.g., LDH, PSA enzymes),Hormonal markers (hCG, calcitonin),Oncofetal antigens (AFP, CEA)

UNIT III

carbohydrate and Glycoprotein Markers,CA-125, CA 15-3, CA 19-9,Clinical relevance and diagnostic challenges,Protein TumorMarkers,PSA and pro-PSA,Thyroglobulin, β2-microglobulin,Serum free light chains,Genetic and Molecular Markers **,**Oncogenes and tumor suppressor genes,BRCA1/2, TP53 alterations,Chromosomaltranslocations,Next-generation sequencing (NGS),Epigenetic markers and methylation profiles

UNIT IV

Laboratory Techniques in Tumor Marker Detection,ELISA, RIA,Westernblotting,PCR, qPCR, RT-PCR,Immunohistochemistry (IHC)

Ethical, Social, and Clinical Issues,Overdiagnosis and overtreatment,Psychological impact of screening,Geneticcounseling and patient privacy,Use of markers in personalized medicine

Practicals

- ELISA for tumor marker detection
- PCR amplification of gene markers
- Flow cytometry basics
- Case-report discussion and result interpretation

Semester VI

1.Course Title :- Microbiology-II

2.Course Code MLMDT 603

3.Typology of Course Core discipline

4.Course Duration/LP (lectures) 90 hours lectures of theory (1 credit = 15 hours of theory learning on 6 Hrs/week

5.Credits 4 credits

6.Max. Marks and Time

External Theory	Internal Theory	External Practical	Internal Practical
40	10	40	10

100Marks (80TH/PR+20IA)

a. The exam of this course will be of 3hours duration.

b. There will be no objective type questions.

c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.

d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.

7.Job role Medical lab technologist

8.Course Objective On successful completion of the course, the Students will be able to :

- To understand basic concepts of microbiology.
- To study microorganisms and their diseases.
- To learn basic microbiological laboratory techniques.

9.Course Outcomes

- Perform basic microbiological test.
- Identify common microorganisms.
- Apply laboratory safety and infection control practices.

10.Pedagogy of the Course work: Class room lecture, laboratory, interactive sessions. PPT, Practical and group discussion

11.Instructions for Paper Setter and candidates:

- The syllabus of this paper has been divided in to FOUR units.
- Examiner will set a total of NINE questions comprising two questions from each unit, including Question no. 1 (compulsory) of short answer type covering the whole syllabus.
- The students are required to attempt one question from each unit and entire Compulsory Question no. 1
- All questions carry equal marks.

Syllabus
MLMDT 603 Microbiology -II
UNIT I

- Preservation of microbes and Lyophilization methods
- Total and viable counts of bacteria
- Testing of disinfectants: Rideal-Walker, Chick-Martin and In-use tests
- Preparation and standardization of vaccines and immunization schedule. Bacteriological examination of water, milk, food and air

UNIT II

- Nosocomial infections and sterility testing of I/V fluids and processing of various samples for hospital infections
- Toxin: Antitoxin assays and pathogenicity tests
- Epidemiological markers of microorganisms: Serotyping, Bacteriophage and Bacteriocin typing methods

- Advanced techniques in microbiology-ELISA, RIA, Co-agglutination GLC, HPLC. Rapid diagnostic methods and Automation in Microbiology

UNIT III

- Handling of fresh histological specimen (tissues), cryo/frozen UNITs of fresh and fixed tissues, freeze drying
- Micro-organisms in tissues, various staining technique for their demonstration
- Cytoplasmic constituents and their demonstration
- Allergy
- Rheumatological diseases and investigations.

UNIT IV

- Advanced techniques and future trends in field of microbiology
- Updates on advanced techniques and future trends in field of diagnostic microbiology
- Updates on advanced techniques and future trends in field of molecular diagnostic technique

Practicals:

1. Reservation of microbes and Lyophilization methods
2. Total and viable counts of bacteria
3. Testing of disinfectants: Rideal-Walker, Chick-Martin and In-use tests
4. Preparation and standardization of vaccines and immunization schedule
5. Eye and Ear fungi infections
6. Serological tests for fungal infections and skin tests
7. Advanced techniques in microbiology-ELISA, RIA, GLC, HPLC

Reference Books

1. Ananthnaryans and Paniker's Text book of Microbiology, CKJ paniker, Orient Longman
2. Roitt's Essential immunology Delves, Peter J., Martin, seamus J. Burton, Dennis R. Roitt, Ivan M Ananthanarayan and Paniker's Textbook of microbiology Kapil , artied
3. Microbiology: An Introduction, Gerard J. TortoraBerdell R. Funke and Christine L. Case
4. Parasitology Chatterjee K.D.
5. Microbiology Pelczar, Michal J and Others
6. Medical Microbiology Greenwood David and Other
7. Ananthanarayan and Panikar's text book of Microbiology Artikapil
8. Immunology Male David and Other
9. Mackie and Mc Cartney Practical Medical Microbiology Collee J.G and Other
10. Bailey and Scott's Diagnostic Microbiology, Patricia Tille
11. Sherris Medical Micrbiology, Pottinger, L. Barth Reller and Charles R. Sterling
12. Stephenson Calculations for Molecular Biology
13. Gerald Karp Cell Molecular Biology
14. Stanier General Microbiology

Semester VI

1.Course Title :- Advance Molecular Biology

2.Course Code MLMDT 604

3.Typology of Course Core discipline

4.Course Duration/LP (lectures) 90 hours lectures of theory (1 credit = 15 hours of theory learning on 6 Hrs/week

5.Credits 4 credits

6.Max. Marks and Time

External Theory	Internal Theory	External Practical	Internal Practical
40	10	40	10

100Marks (80TH/PR+20IA)

a. The exam of this course will be of 3hours duration.

b. There will be no objective type questions.

c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.

d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.

7.Job role Medical lab technologist

8.Course Objective On successful completion of the course, the Students will be able to :

- To understand advanced concept of molecular biology.
- To learn modern molecular techniques and their applications.
- To understand molecular mechanism of gene expression and regulation.

9.Course Outcomes

- Students will able to perform basic molecular techniques .
- Students will be able to analyze DNA ,RNA and gene expression.
- Students will be able to apply molecular methods in disease diagnosis.

10.Pedagogy of the Course work: Class room lecture, laboratory, interactive sessions. PPT, Practical and group discussion

11.Instructions for Paper Setter and candidates:

- The syllabus of this paper has been divided in to FOUR units.
- Examiner will set a total of NINE questions comprising two questions from each unit, including Question no. 1 (compulsory) of short answer type covering the whole syllabus.
- The students are required to attempt one question from each unit and entire Compulsory Question no. 1
- All questions carry equal marks.

Syllabus

MLMDT 604 Advanced Molecular biology

UNIT I

- Introduction: Role of molecular diagnostics in present diagnostic area, Benefits of molecular diagnostics over serological diagnostic tests, Ethical issues related to molecular diagnostics, Basic techniques used in molecular diagnostics, future of molecular diagnostics

UNIT II

- Sophisticated Techniques: DNA profiling, blotting techniques, VNTRs, RFLP, RAPD
- Detection of gene mutation- Hybridization-Based Methods, Sequencing (Polymerization)-Based Methods, Cleavage Methods, Detection of SNPs using agarose based and PCR based methods

UNIT III

- Molecular diagnostic of various viral diseases: HIV type -I, HIV type –II, HPV, Herpes, Various hepatitis strains, Influenza (H1N1), sample preparation, various steps required for viral infection analysis and Viral load monitoring
- Molecular diagnostics of bacterial infections: Mycobacterium tuberculosis, Pathogenic E Coli, sample preparation and pathogen detection, parasitic diseases – Neiseria gonorrhoeae, malaria

UNIT IV

- Genetic disorders and inborn errors of metabolism: Monogenetic disorder – e.g. Cystic fibrosis etc., Epigenetic disorder – e.g. Cancer etc., Polygenetic disorder – e.g. Diabetes, Triple repeat, Obesity, etc. Inborn error of metabolism: Lipidosis, Lysosomalstorage disorders, glycogen storage disorders, Gaucher and Pompe, mucopolysaccharidoses- Hunter and Hurler, phenyl ketonuria, alkaptonuria, tyrosinosis
- Neuronal Defects: Haemoglobinopathies, Neuro-degenerative disorders- Parkinson's, Alzheimer and Meningitis

Practical:

1. Agarose Gel Electrophoresis
2. Polyacrylamide Gel Electrophoresis
3. PCR
4. Western & Southern Blotting
5. Gradient Gel Electrophoresis

Reference Books:

1. Rao, J.R., Fleming, C.C., & Moore, J.E., Molecular Diagnostics: Current Technology and Applications, Colin Creaig Fleming, Horizon Scientific Press
2. M., Narosa, Radhakrishnan, S., Patil, K.M., Medical Diagnostics and Procedures, Singh, Publishing House
3. Scot, W.K., & Ritchie, M.D., Genetic Analysis of Complex Disease John Willey
4. Homes, F.A., Techniques in diagnostic Human Biochemical Genetics, Wiley-Blackwell

Semester VI

1.Course Title :- Metabolic Biochemistry

2.Course Code MLMBT-603

3.Typology of Course Core discipline

4.Course Duration/LP (lectures) 90 hours lectures of theory (1 credit = 15 hours of theory learning on 6 Hrs/week

5.Credits 4 credits

6.Max. Marks and Time

External Theory	Internal Theory	External Practical	Internal Practical
40	10	40	10

100Marks (80TH/PR+20IA)

a. The exam of this course will be of 3hours duration.

b. There will be no objective type questions.

c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.

d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.

7.Job role Medical lab technologist

8.Course Objective On successful completion of the course, the Students will be able to :

- To understand major metabolic pathways.
- To study regulation of carbohydrate, protein, and lipid metabolism.
- To understand metabolic disorders and their biochemical basis.

9.Course Outcomes

- Students will be able to explain major metabolic pathways.
- Students will be able to relate metabolism to disease conditions.
- Students will be able to interpret basic biochemical finding Students will safely handle specimens, chemicals, and laboratory instruments..

10.Pedagogy of the Course work: Class room lecture, laboratory, interactive sessions. PPT, Practical and group discussion

11.Instructions for Paper Setter and candidates:

- The syllabus of this paper has been divided in to FOUR units.
- Examiner will set a total of NINE questions comprising two questions from each unit, including Question no. 1 (compulsory) of short answer type covering the whole syllabus.
- The students are required to attempt one question from each unit and entire Compulsory Question no. 1
- All questions carry equal marks.

Syllabus
MLMDT 605 METABOLIC BIOCHEMISTRY
Unit I:

Fundamentals of Metabolism & Bioenergetics

- Introduction to metabolism: definition, scope, significance in living systems; catabolism vs anabolism; cellular compartments of metabolic pathways.
- Energy currency of the cell: ATP, other high-energy compounds; thermodynamics of biochemical reactions: free energy, ΔG, endergonic vs exergonic reactions.
- Redox reactions, cofactors and coenzymes (NAD^+/NADH, NADP, FAD, etc.), reducing equivalents in metabolism .
- Regulation of metabolic pathways: allosteric regulation, hormonal regulation, feedback mechanisms, integration between pathways (homeostasis) .

Unit II

Carbohydrate Metabolism

- Digestion, absorption and assimilation of carbohydrates (overview) .
- Key metabolic pathways: Glycolysis, Gluconeogenesis, Pentose Phosphate Pathway, TCA (Krebs) cycle, Glycogen synthesis (glycogenesis) & Glycogen breakdown (glycogenolysis), integration of pathways.
- Regulation of carbohydrate metabolism: hormonal regulation (e.g. insulin, glucagon), allosteric regulation; energy balance in fed vs fasting state.

Unit III

Lipid, Protein/Amino-acid and Nucleotide Metabolism

- Lipid metabolism: digestion/absorption of lipids (overview), mobilization of fats, transport (lipoproteins), β-oxidation of fatty acids (saturated & unsaturated)..
- Protein / amino-acid metabolism: digestion & absorption of proteins; amino-acid catabolism (transamination, deamination, decarboxylation), urea cycle, nitrogen balance, disposal of nitrogen, metabolic fate of ammonia.
- Nucleotide (Purine & Pyrimidine) metabolism: biosynthesis (de novo and salvage pathways), degradation, regulation; clinical relevance (e.g. hyperuricemia, gout, metabolic disorders).

Unit IV

Integration of Metabolism, Bioenergetics & Clinical Correlations

- Integration of metabolic pathways: interconnection among carbohydrate, lipid, amino-acid, nucleotide metabolism; how metabolism adapts during fed, fasting, starvation states; tissue/organ-specific metabolism (liver, adipose, muscle, brain) — especially relevant for lab diagnostics.
- Bioenergetics and oxidative metabolism: electron transport chain (ETC), oxidative phosphorylation, ATP yield, regulation and inhibitors, coupling/uncoupling — core to energy metabolism.

- Metabolic disorders & biochemical abnormalities: inborn errors of metabolism, disorders of carbohydrate/lipid/protein metabolism.

Practicals

Practical Syllabus:

1. Qualitative tests for carbohydrates

Molisch's, Benedict's, Fehling's, Barfoed's, Seliwanoff's, Iodine test.

2. Estimation of Blood Glucose

a. GOD–POD method

b. Colorimetric estimation using standard graph.

3. Estimation of Glycogen from tissue extract

Acid hydrolysis method (anthrone reagent).

4. Glucose Tolerance Test (GTT) – Demonstration

Calculation and interpretation of glycemic curves.

5. Qualitative tests for lipids

Solubility, emulsification, saponification, acrolein test.

6. Qualitative tests for proteins and amino acids

Biuret, Ninhydrin, Xanthoproteic, Millon's, Hopkins–Cole.

7. Estimation of Uric Acid

Uricase-peroxidase method.

8. Analysis of amino acids using Paper Chromatography

Identification of amino acids by Rf values.

9. Assay of Amylase activity

Salivary amylase with starch substrate; iodine color reduction.

10. Assay of Lipase

Preparation of calibration curves, internal QC, Levey–Jennings chart.

Recommended resources:

1. Textbook of Biochemistry for Medical Students – D.M. Vasudevan, Sreekumari & Vaidyanathan
2. Biochemistry – U. Satyanarayana&Chakrapani
3. Essentials of Biochemistry – Debajyoti Das / S. Mukherjee
4. Biochemistry for Medical Students – P.C. Pattanayak
5. Fundamentals of Biochemistry – J.L. Jain
6. Textbook of Medical Biochemistry – M.N. Chatterjee & Rana Shinde
7. Clinical Biochemistry – D.M. Vasudevan& S. Sreekumari
8. Lehninger Principles of Biochemistry – David L. Nelson & Michael M. Cox
9. Harper's Illustrated Biochemistry – Victor W. Rodwell et al.
10. Biochemistry – Jeremy M. Berg, John Tymoczko&LubertStryer
11. Biochemistry – Donald Voet& Judith Voet
12. Medical Biochemistry – John T. Gallagher & David A. Bender
13. Clinical Biochemistry: Metabolic and Clinical Aspects – William J. Marshall &Márta Lapsley
14. Clinical Chemistry: Principles, Techniques, and Correlations – Michael L. Bishop
15. Principles and Techniques of Biochemistry and Molecular Biology – Wilson & Walker

Semester VI

1.Course Title :- Bioinformatics

2.Course Code MLMDT 606

3.Typology of Course Core discipline

4.Course Duration/LP (lectures) 90 hours lectures of theory (1 credit = 15 hours of theory learning on 6 Hrs/week

5.Credits 4 credits

6.Max. Marks and Time

External Theory	Internal Theory	External Practical	Internal Practical
40	10	40	10

100Marks (80TH/PR+20IA)

a. The exam of this course will be of 3hours duration.

b. There will be no objective type questions.

c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.

d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.

7.Job role Medical lab technologist

8.Course Objective On successful completion of the course, the Students will be able to :

- To understand basic concepts and applications of bioinformatics.
- To learn biological databases and sequence analysis.
- To understand computational tools used in molecular biology.

9.Course Outcomes

- Students will be able to use biological databases effectively.
- Students will be able to analyze DNA and protein sequences.
- Students will be able to apply bioinformatics tools in researh.

10.Pedagogy of the Course work: Class room lecture, laboratory, interactive sessions. PPT, Practical and group discussion

11.Instructions for Paper Setter and candidates:

- The syllabus of this paper has been divided in to FOUR units.
- Examiner will set a total of NINE questions comprising two questions from each unit, including Question no. 1 (compulsory) of short answer type covering the whole syllabus.
- The students are required to attempt one question from each unit and entire Compulsory Question no. 1
- All questions carry equal marks.

Syllabus

MLMDT 606 BIOINFORMATICS

UNIT I

Introduction to Bioinformatics,What is bioinformatics?,Applications in medicine, agriculture, and research,Overview of biological data (DNA, RNA, proteins), Biological Databases,What are databases?,Using NCBI, UniProt, and PDB.How to search for a gene or protein

UNIT II

DNA and Protein Sequences,Structure of DNA and proteins (simple review),Understanding FASTA format,Viewing sequences online,Sequence Alignment Basics,What is alignment and why is it used?,Using BLAST (online version only),Interpreting simple BLAST results

UNIT III

What are genes and Genomes, Introduction to genomes and genome browsers, Exploring a genome on UCSC or Ensembl (guided activity)Information and Functions, Protein structure overview, Searching proteins in UniProt

UNIT IV

Exploring simple protein structure models, Phylogenetic Trees Basics ,What is evolution in bioinformatics? Simple phylogenetic

tree concepts Online tree-building tools (very basic),
Applications in Health and Medicine

Practicals

Introduction to Biological Databases
Exploring Protein Structure
Working With DNA Sequences
Basic BLAST Search
Multiple Sequence Alignment (MSA)
Protein Information Search
Exploring Protein Structure

Recommended resources:

NCBI website,UniProt,UCSC Genome Browser,EMBL-EBI training videos,Khan Academy Biology (for biological basics),No textbook required, but optional:,"Introduction to Bioinformatics" by Arthur Lesk (very beginner-friendly)

Semester VI

1.Course Title:- Sensitization to blood bank and infection control

2. Course Code MLMDT 607
3.Typology of Course Core discipline
4.Course Duration/LP (lectures) 90 hours lectures of theory (1 credit = 15 hours of theory learning on 6 Hrs/week

5.Credits 4 credits
6.Max. Marks and Time

External Theory	Internal Theory	External Practical	Internal Practical
80	20	---	---

100Marks (80TH/PR+20IA)
a. The exam of this course will be of 3hours duration.
b. There will be no objective type questions.
c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees.

7.Job role medical lab technologist
8.Course Objective On successful completion of the course, the Students will be able to :

- To understand basic blood banking principles.
- To learn blood grouping and compatibility testing.
- To understand safe blood collection and storage.

9.Course Outcomes

- Students will be able to perform basic blood grouping tests.
- Students will understand blood transfusion safety practices.
- Students will be able to follow basic blood bank procedures. Students will safely handle specimens, chemicals, and laboratory instruments.

10.Pedagogy of the Course work: Classroom lecture, laboratory, interactive sessions. PPT, Practical and group discussion

11.Instructions for Paper Setter and candidates:

- The syllabus of this paper has been divided in to FOUR units.
- Examiner will set a total of NINE questions comprising two questions from each unit, including Question no. 1 (compulsory) of short answer type covering the whole syllabus.

- The students are required to attempt one question from each unit and entire Compulsory Question no. 1
- All questions carry equal marks.

Syllabus

MLMDT 607 Sensitization to Blood Banking and Infection Control

UNIT I

- Components of blood
- Immuno-hematology in detail
- ABO blood group system in detail
- Rh blood group system in detail

UNIT II

- Methodology to identify blood groups
- Different aspects of Blood transfusion techniques
- Investigation of transfusion reaction
- Transfusion of various components of blood

UNIT III

- Serum immunoglobulin
- Different aspects of working in blood
- Storage of Blood Bank

UNIT IV

- Infection control and prevention:
- Practices to curb infection
- Hospital borne infections
- Prevention and treatment of needle stick injury
- Management of blood and body substance spills in the health care setting
- The path of disease transmission
- paths of transmission including direct contact and penetrating injuries
- risk of acquisition
- sources of infecting microorganisms including persons who are carriers
- The incubation phase of the disease or those who are acutely ill
- Aspects of infectious diseases including- opportunistic organisms, pathogens

Reference Books

1. Atlas of Haematology, G.A. McDonald
2. Clinical Haematology, Christopher A. Ludlam
3. Practical Haematology, J.B. Dacie
4. Practical Haematology , Sir John
5. Haematology, Emmanuel C.Besa
6. Haematology, Pathophysiological basis for clinical practice, Stephen M. Robinson
7. Haematology for students Practitioners, RamnikSood
8. Hand book of Medical Laboratory Technology, V.H. Talib
9. Handbook of Blood Banking and Transfusion Medicine, Rao Gundu HR, JagannathanLatha, Eastlund Ted
10. Modern Blood Banking & Transfusion Practices Hardcover, Denise M Harmening
11. Textbook of Blood Banking and Transfusion Medicine, Elsevier eBook on VitalSource, Sally V. Rudmann,
12. Textbook of Blood Banking and Transfusion Medicine, Sally V. Rudmann
13. Hospital Epidemiology and Infection Control , C. Glen Mayhall

14. Hospital Acquired Infections: Prevention and Control Paperback – Import, PurvaMathur
15. Hospital Infection Control Guidelines: Principles and Practice, Singh Sanjeev, Gupta Shakti Kumar, Kant Sunil
16. Bennett & Brachman's Hospital Infections, William R. Jarvis MD